Fundamental limits for thermodynamic control with quantum feedback

Kaiyuan Ji (Cornell), Gilad Gour (Technion), Mark M. Wilde (Cornell)

arXiv:2503.09012

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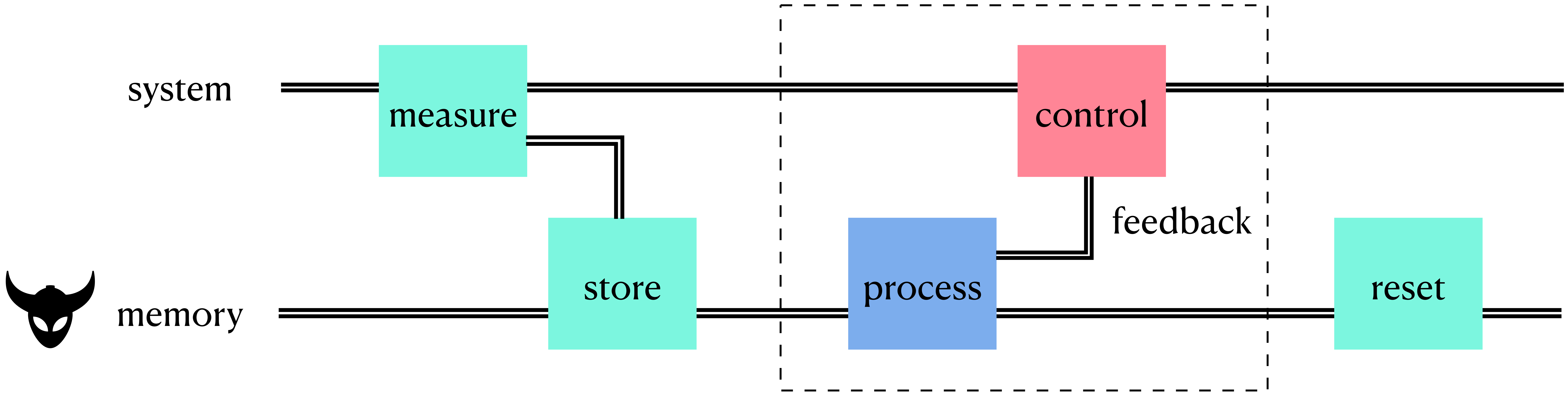

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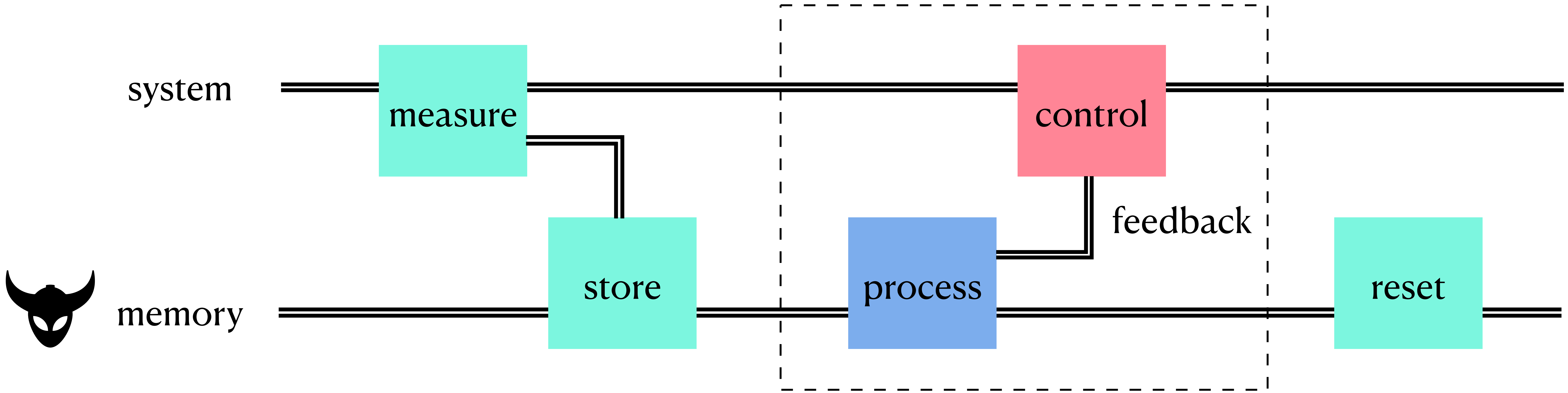

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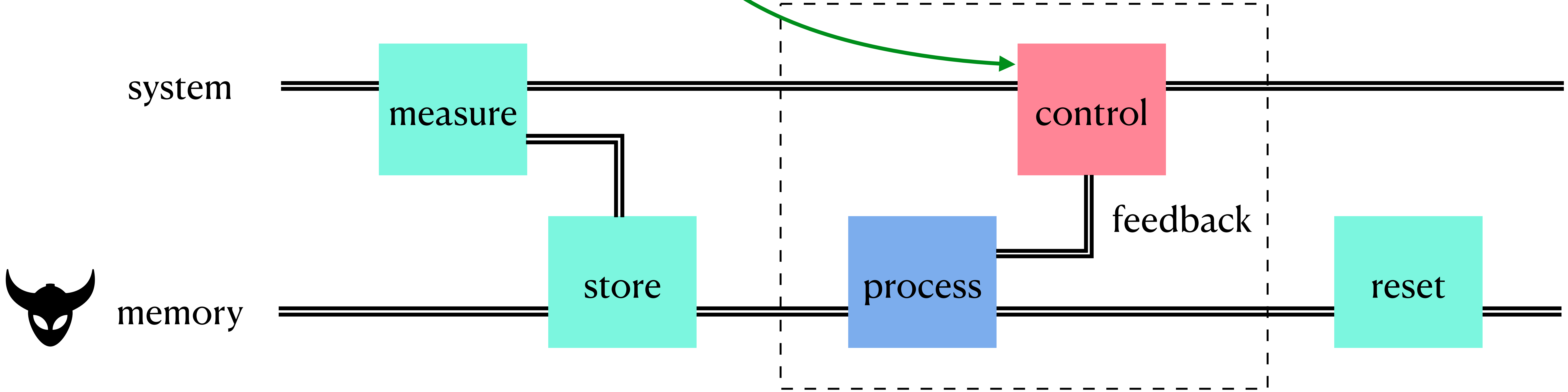

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feedback advantage would be paid back in measurement & reset

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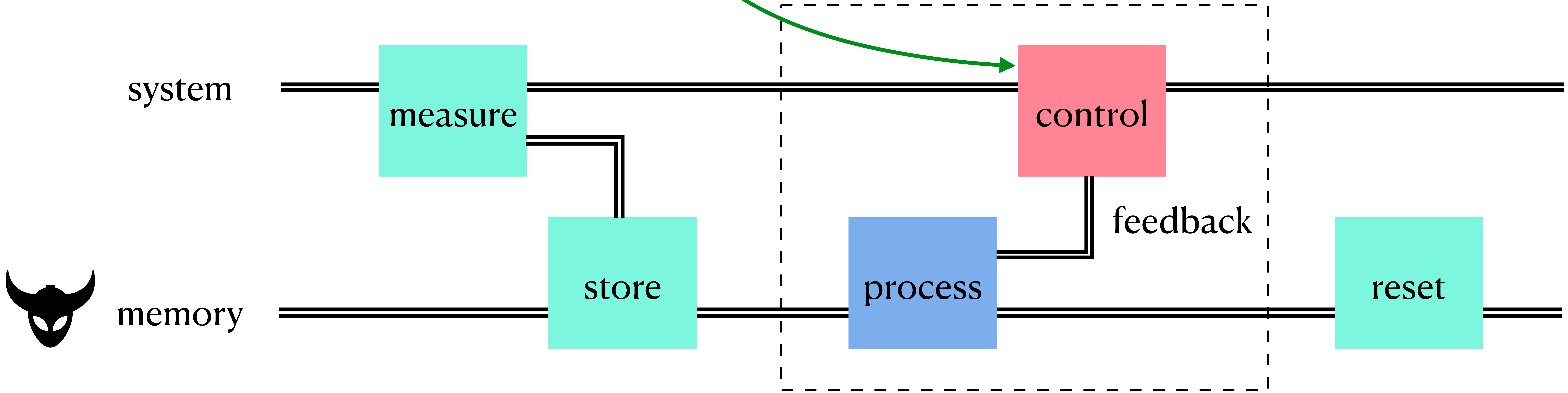

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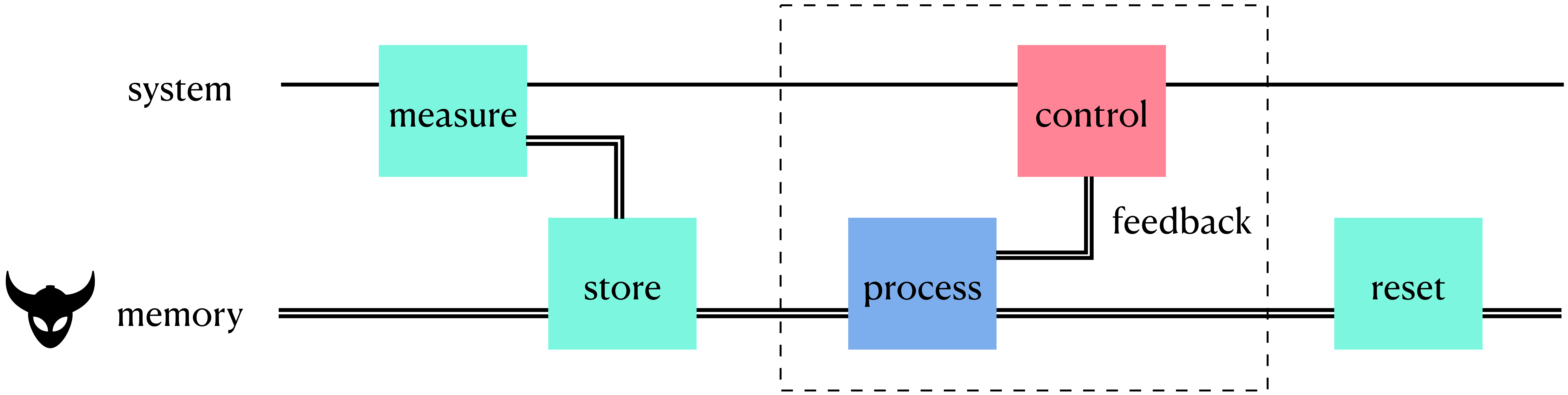

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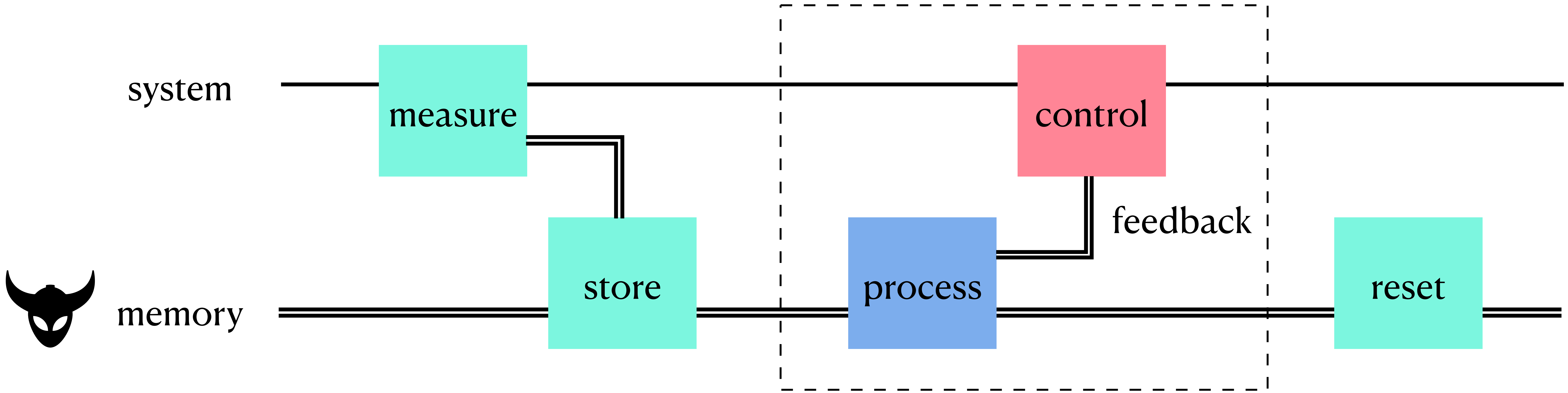

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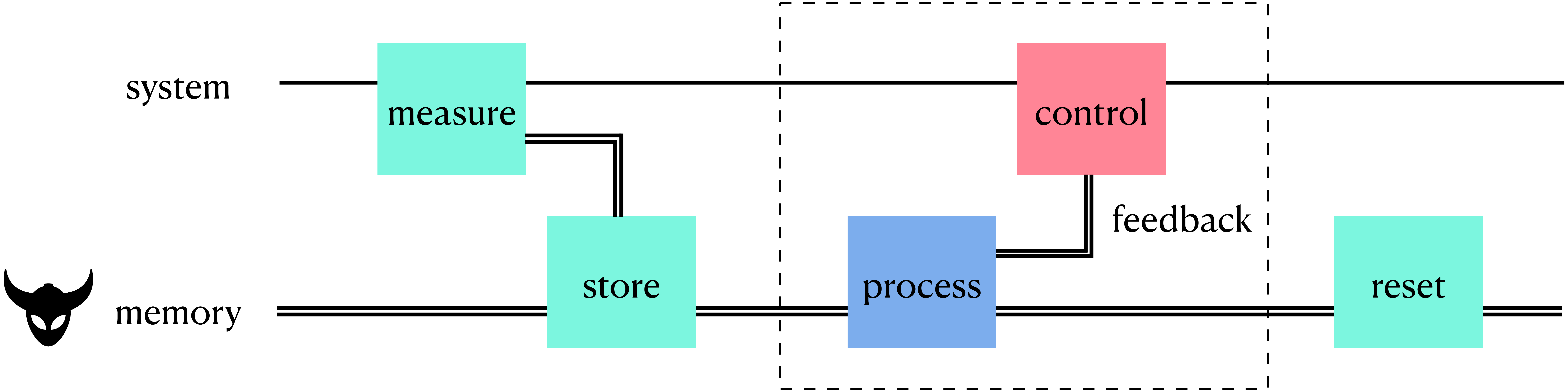
- Even weak or indirect measurements on quantum systems can only yield **classical** side information, which is then fed back classically



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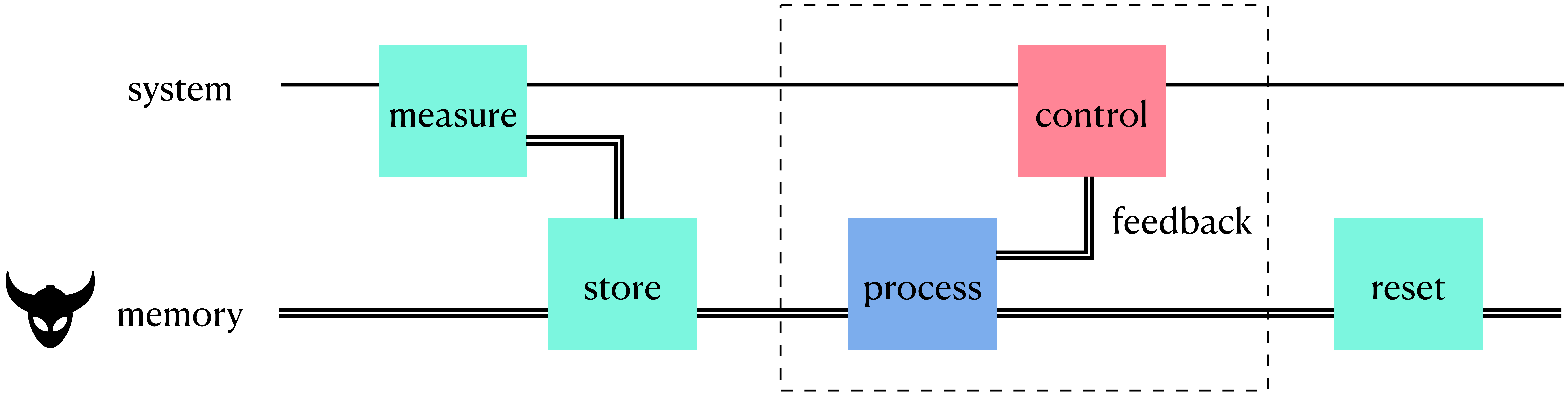
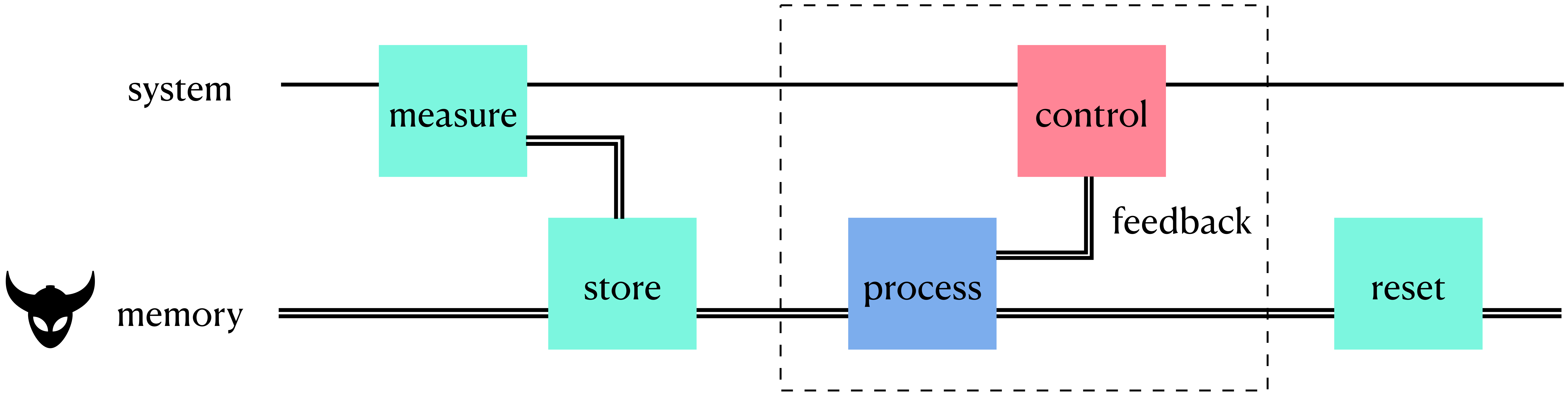
Generalizations to quantum systems & control were also studied

- But side information is always assumed to be collected by measurements
- Even weak or indirect measurements on quantum systems can only yield **classical** side information, which is then fed back classically
- Such assumptions do not capture the most general situation allowed by physics



Fully quantum feedback

A fully quantum feedback-controller would be able to acquire, process, and feed back **quantum** side information coherently



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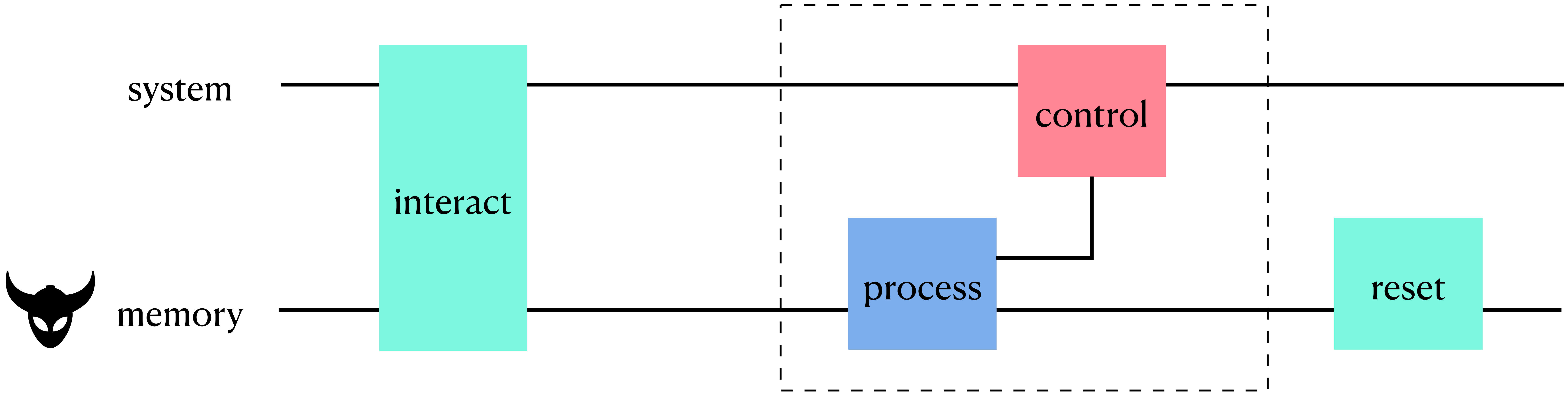

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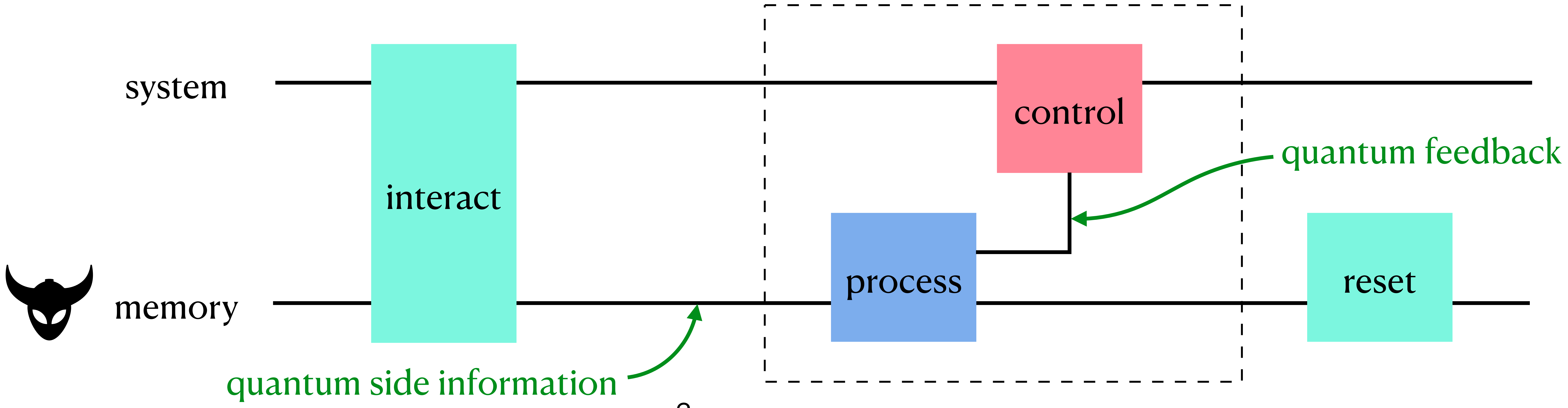

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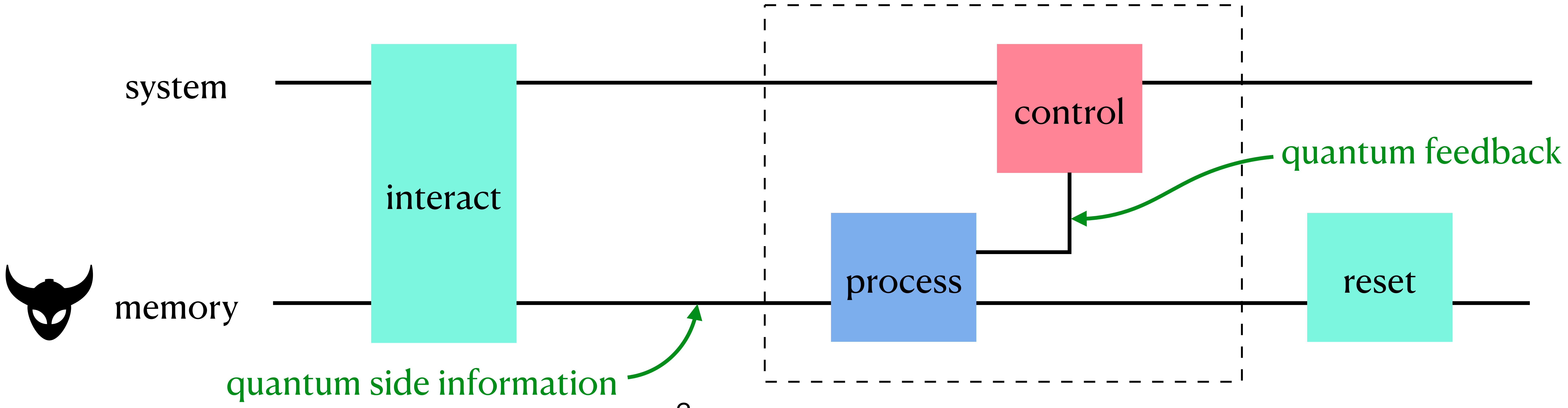
Seth Lloyd, PRA 62, 022108 (2000)

- Some tasks in quantum control theory, such as state exchange and entanglement transfer, are only achievable with quantum feedback; impossible with classical feedback
- Only bounds assuming quantum feedback qualify as “fundamental no-go limits” for all physically admissible feedback-control processes



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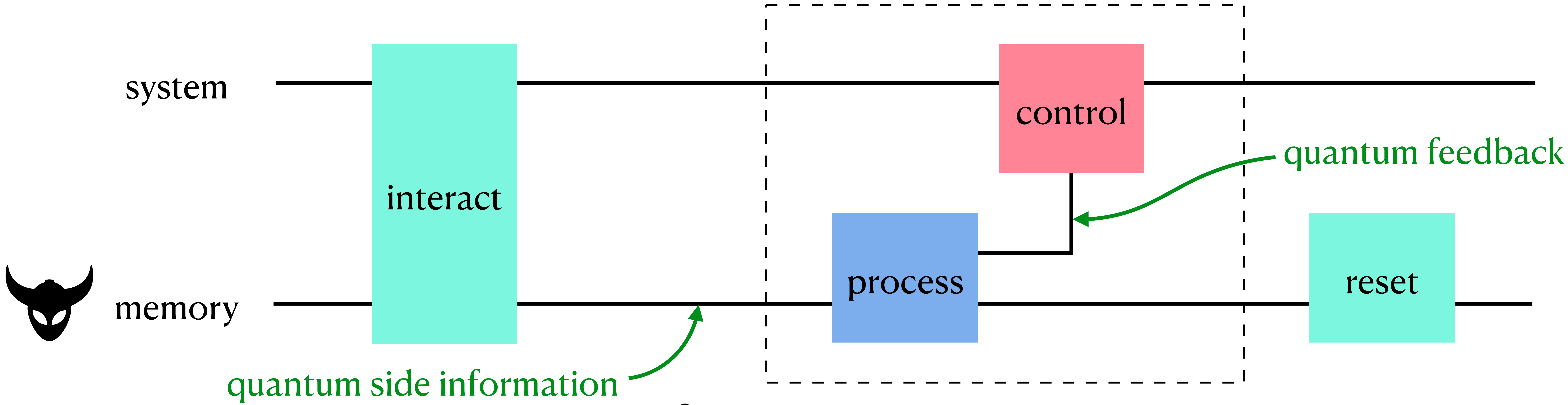

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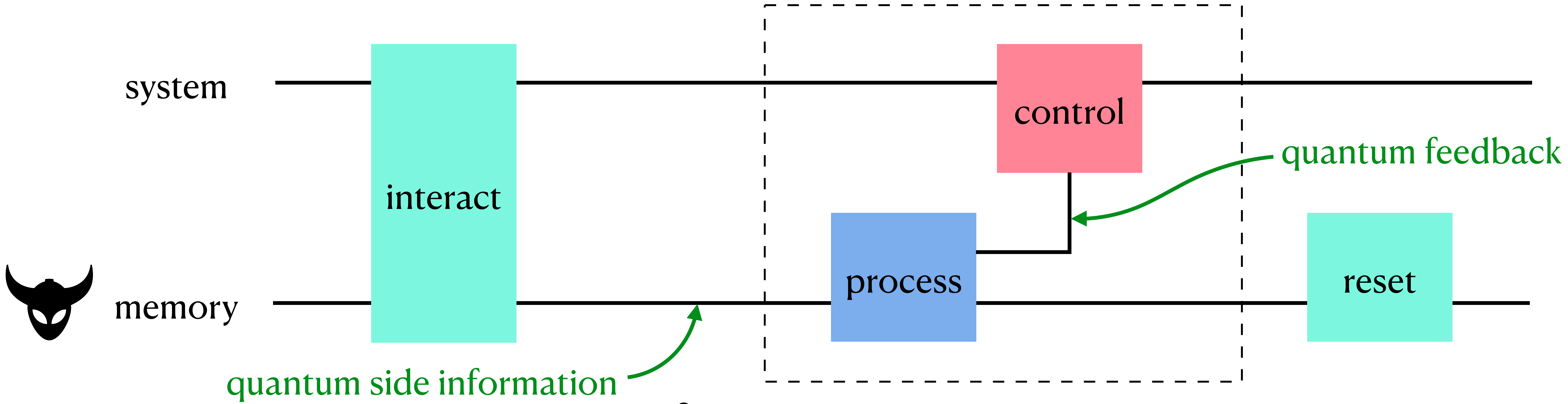
del Rio et al., Nature 474, 61 (2011)

- May lead to exotic thermodynamic effects (such as cooling during erasure) due to negativity of quantum conditional entropy



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- under fully quantum feedback control
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Need a precise description for microscopic thermodynamic operations with quantum feedback

- Motivates us to define a resource theory of thermodynamics with quantum side information

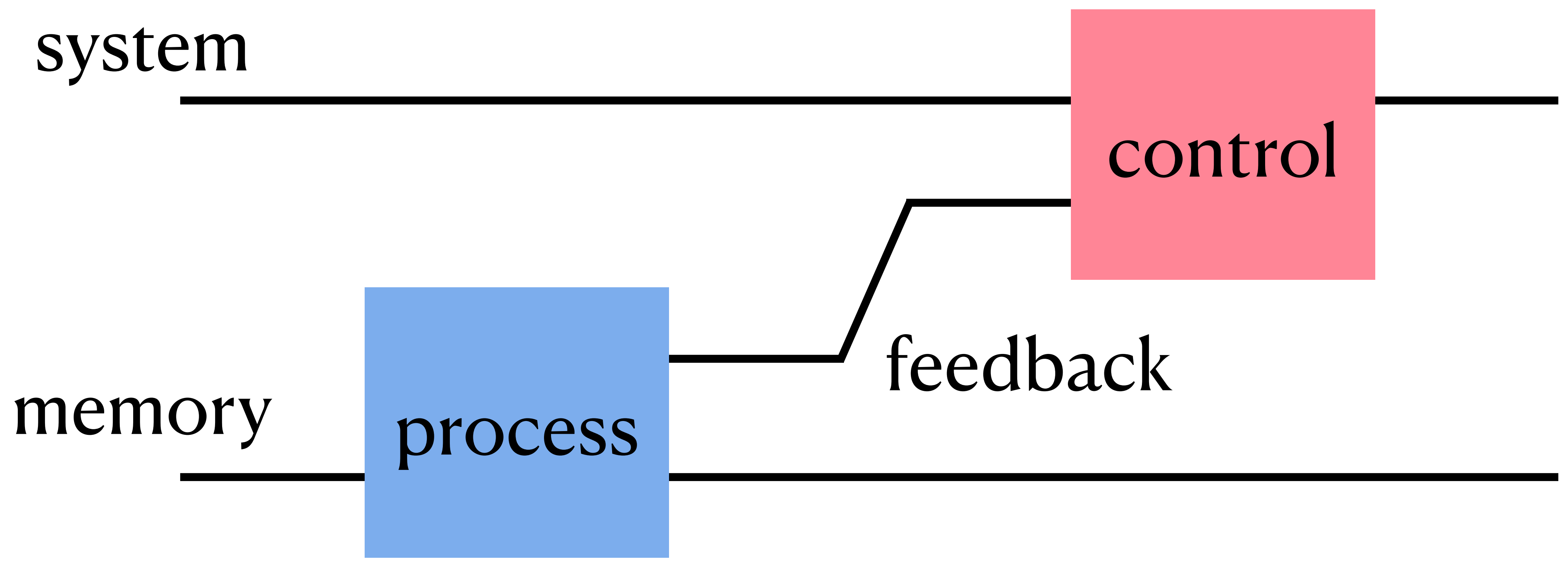
Operational framework:

Q. thermodynamics in the presence of q. side information

Free operations

We only focus on feedback control per se

Everything before that (e.g., acquisition of q. side info.) is encapsulated in the input state



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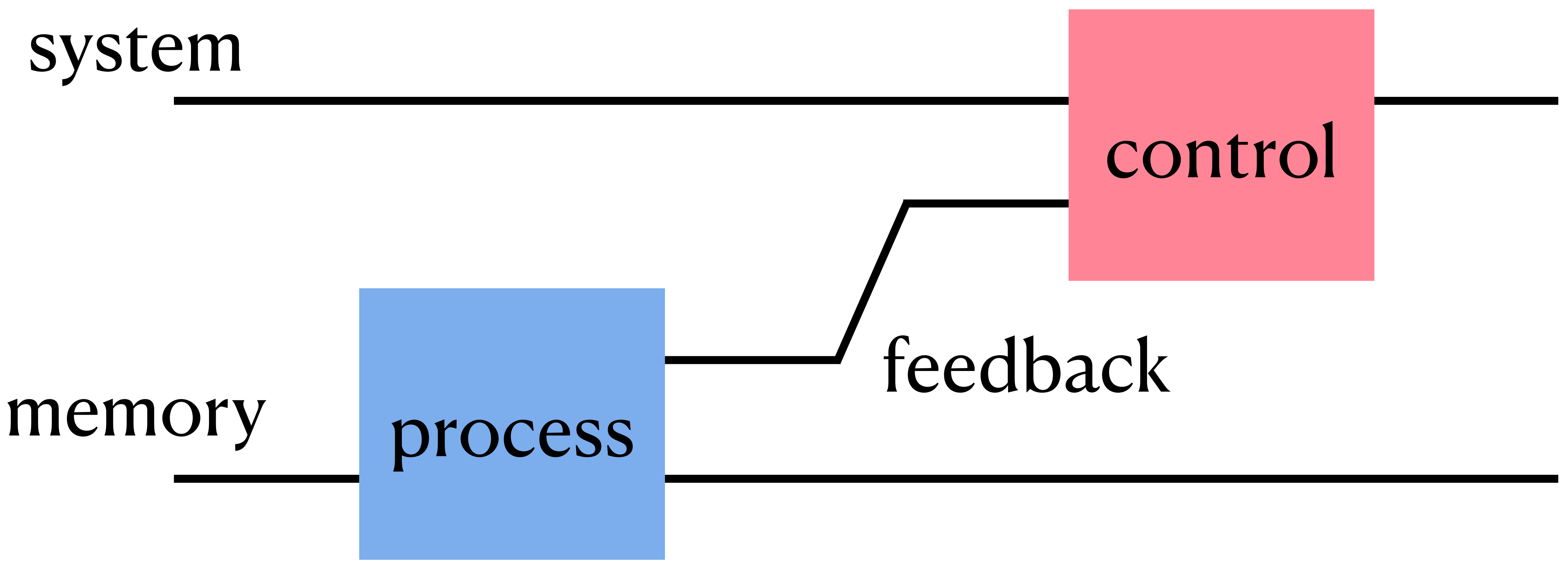
Axiomatic approach: every physically admissible feedback-control scheme that can be implemented without supplying work to the system should be free



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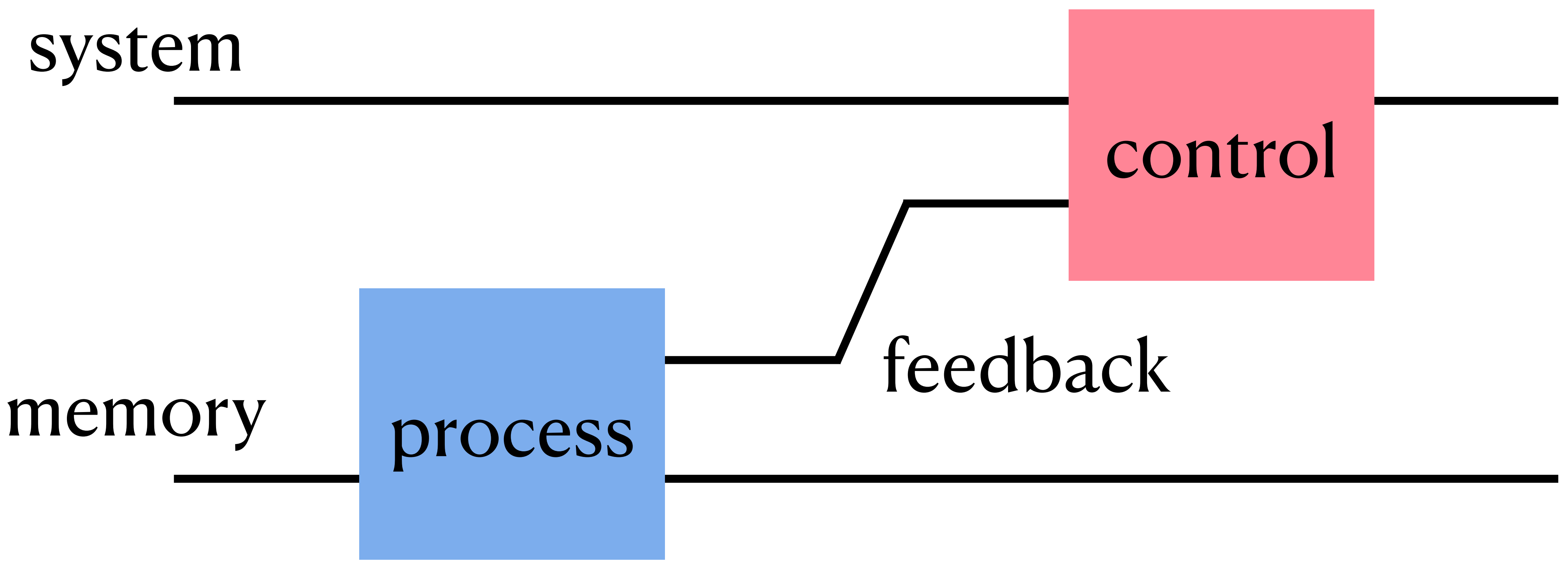

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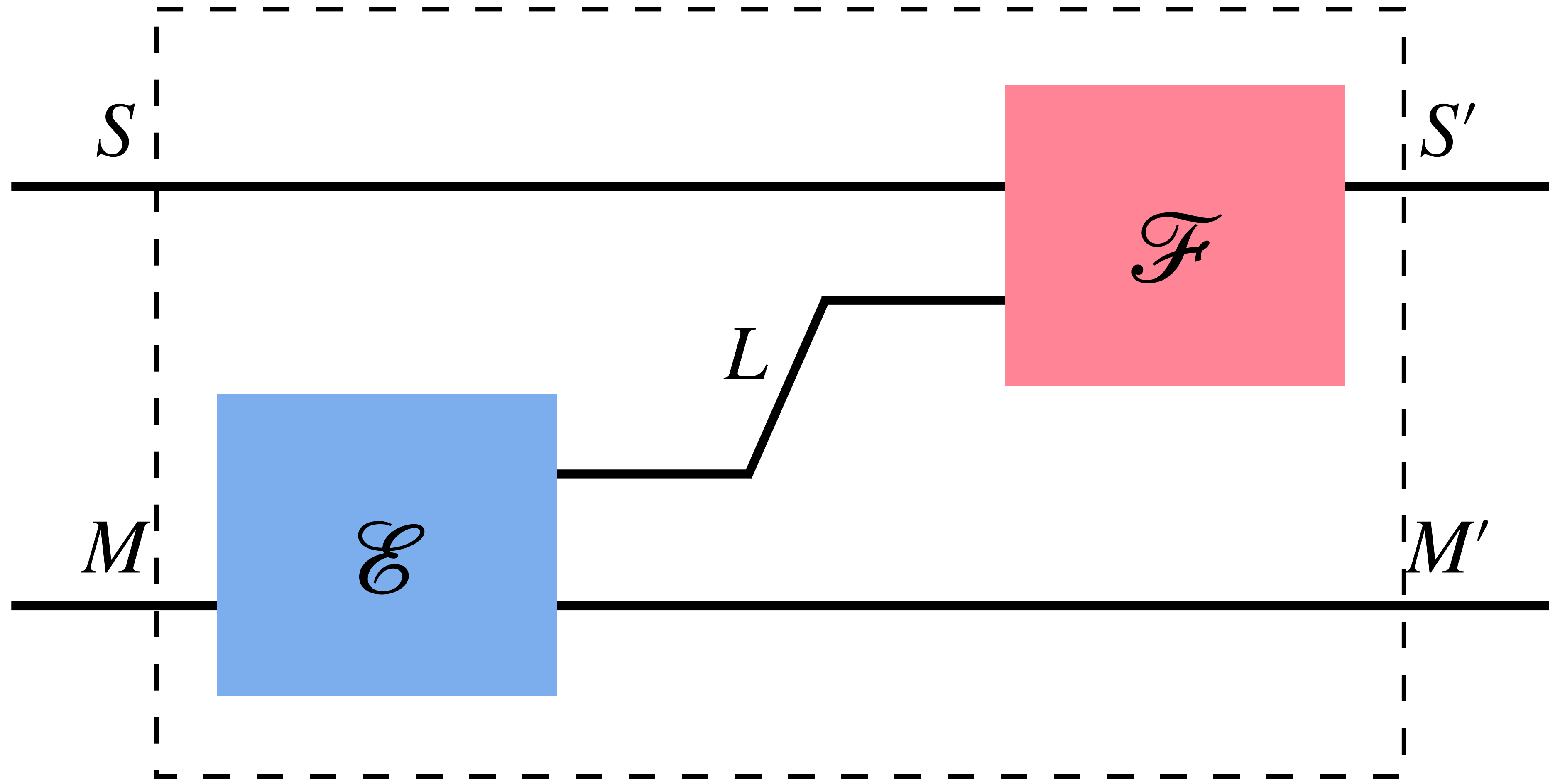

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L is a quantum register

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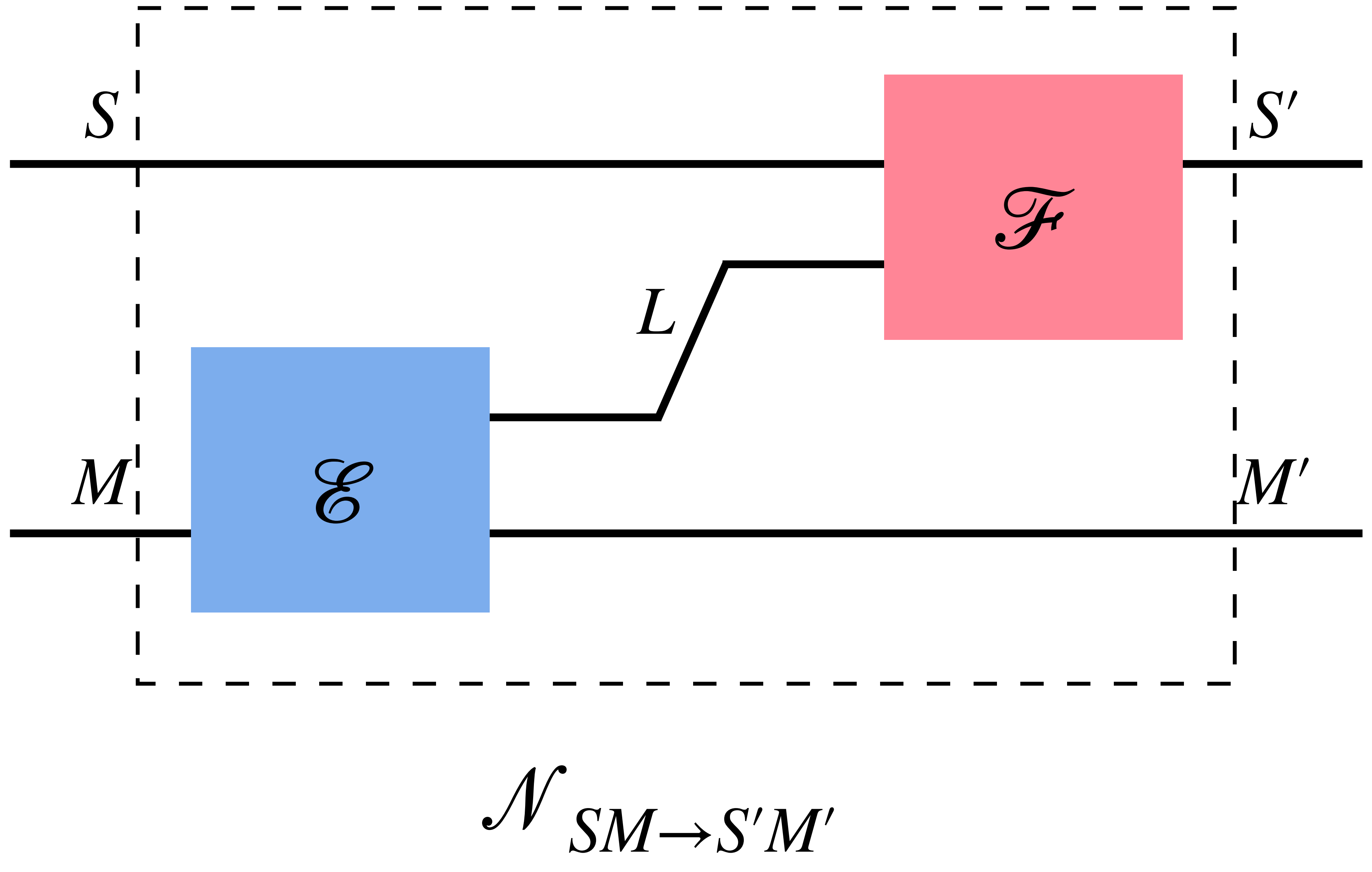

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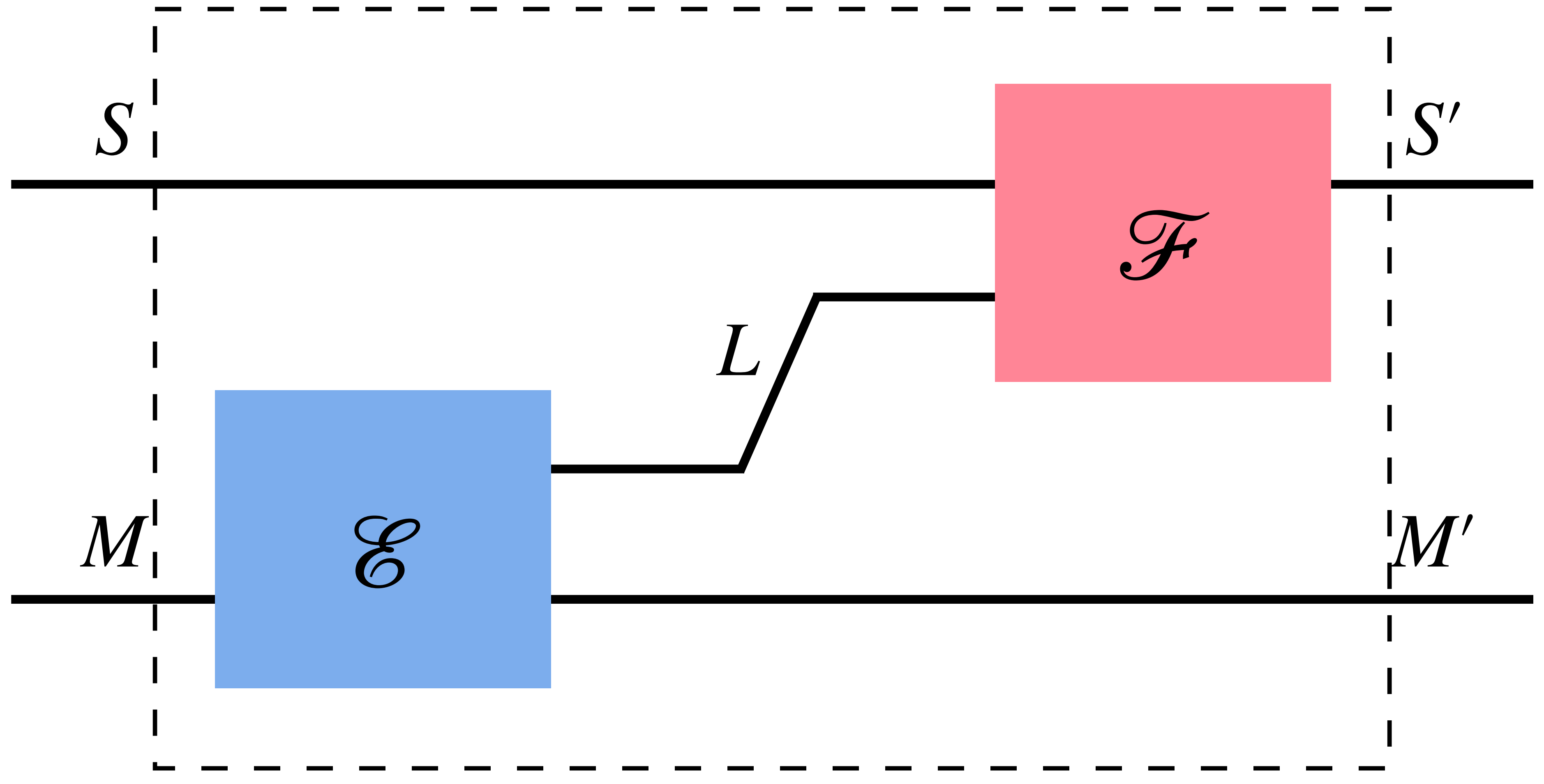

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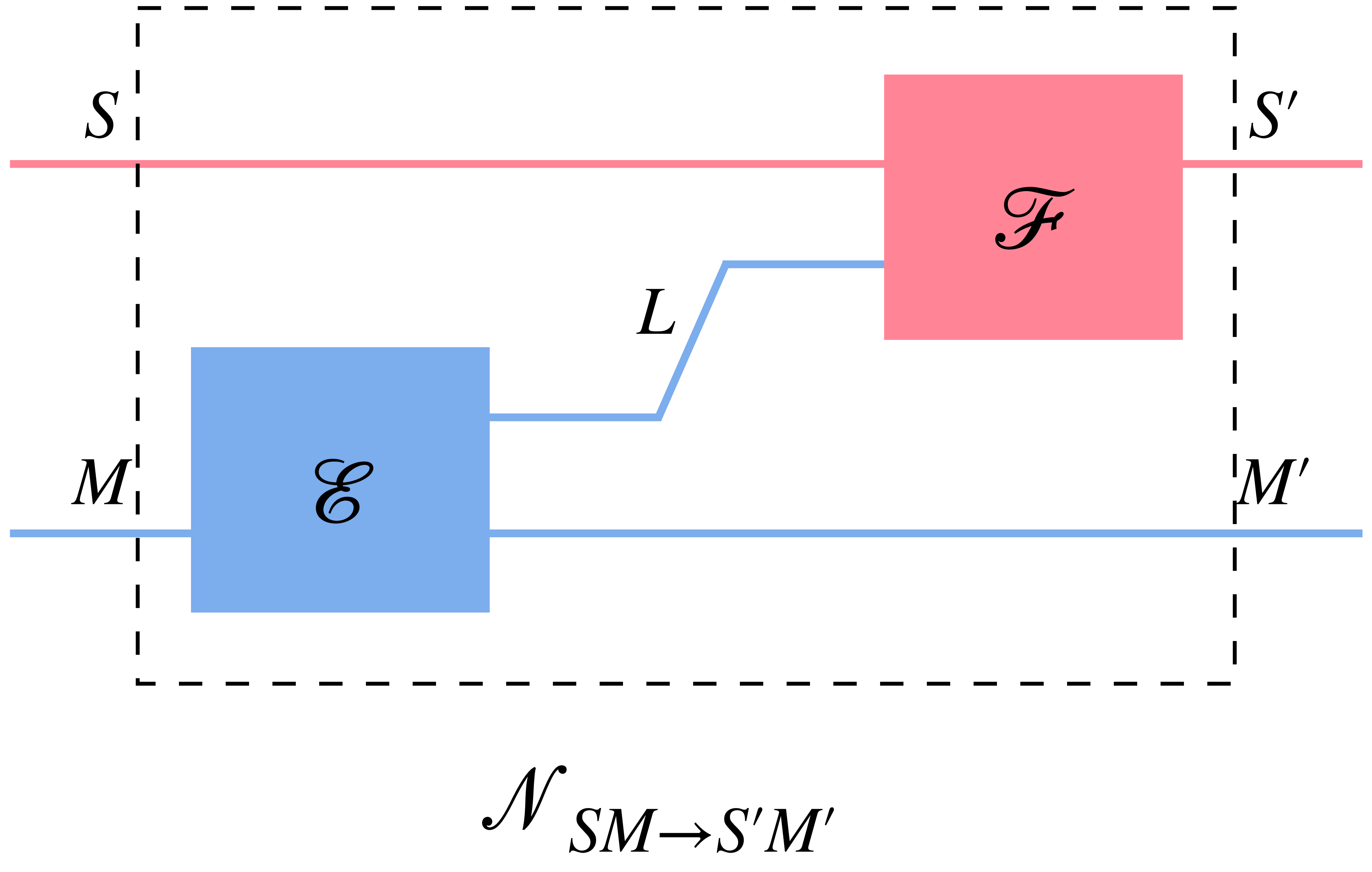

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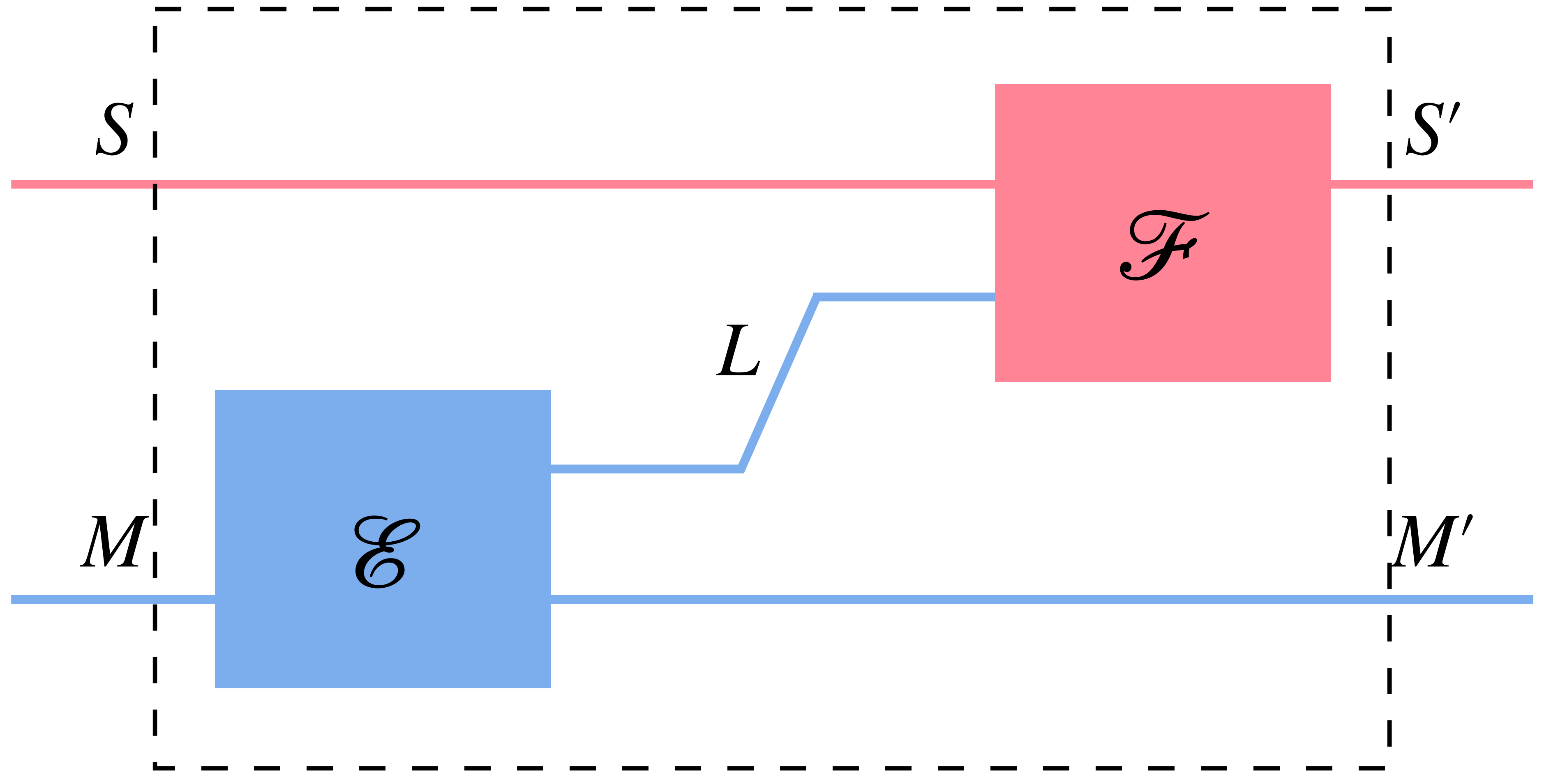

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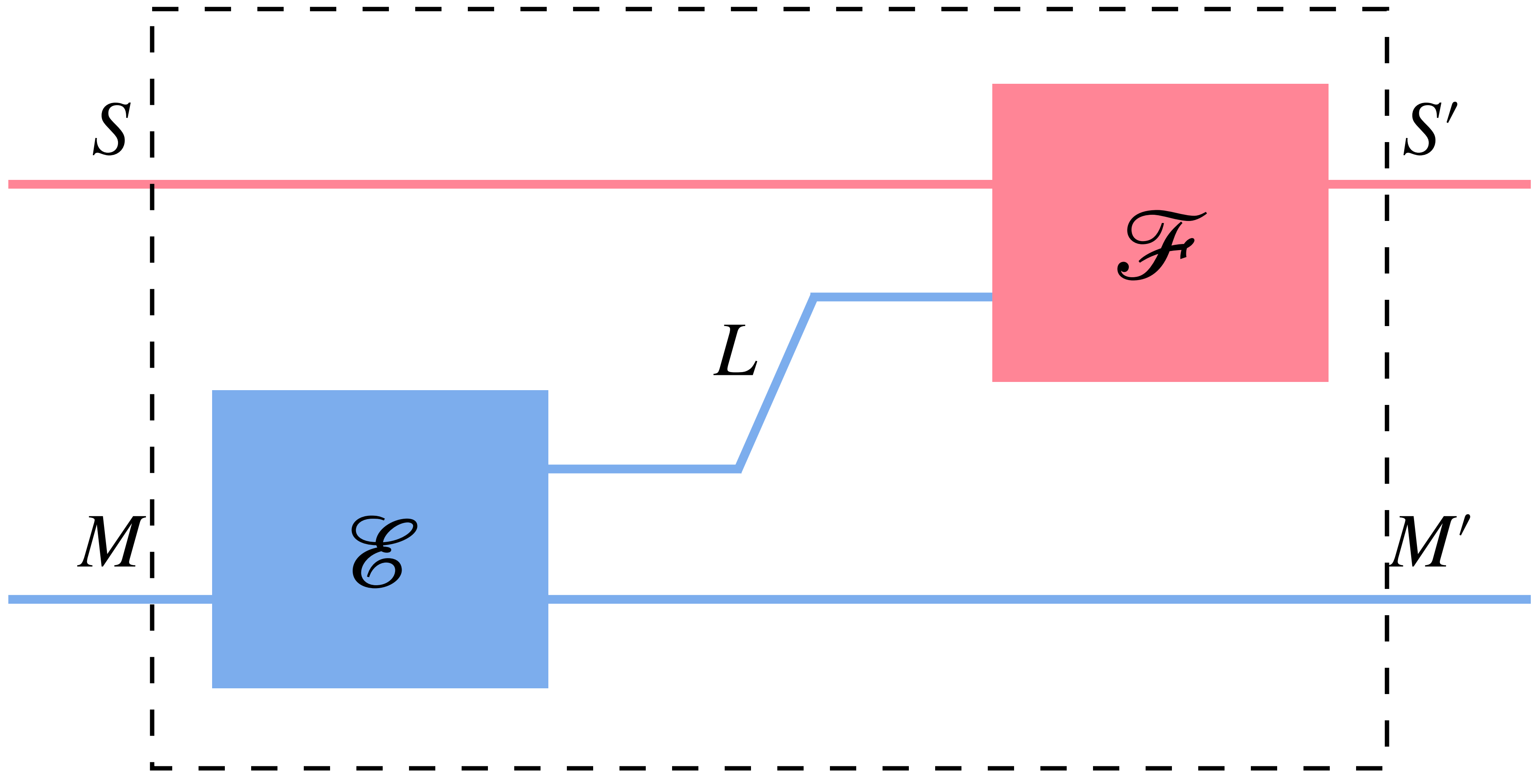

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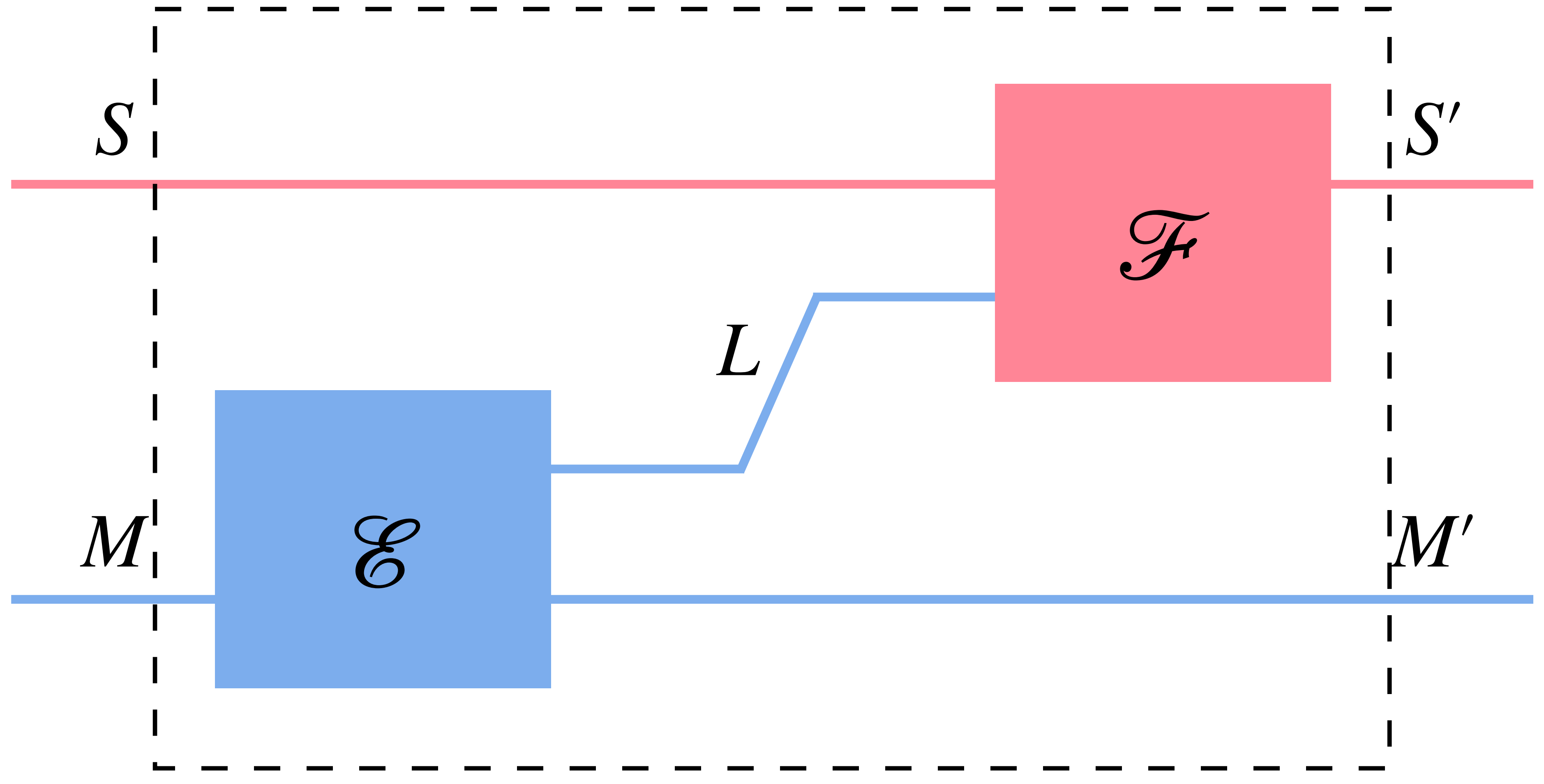

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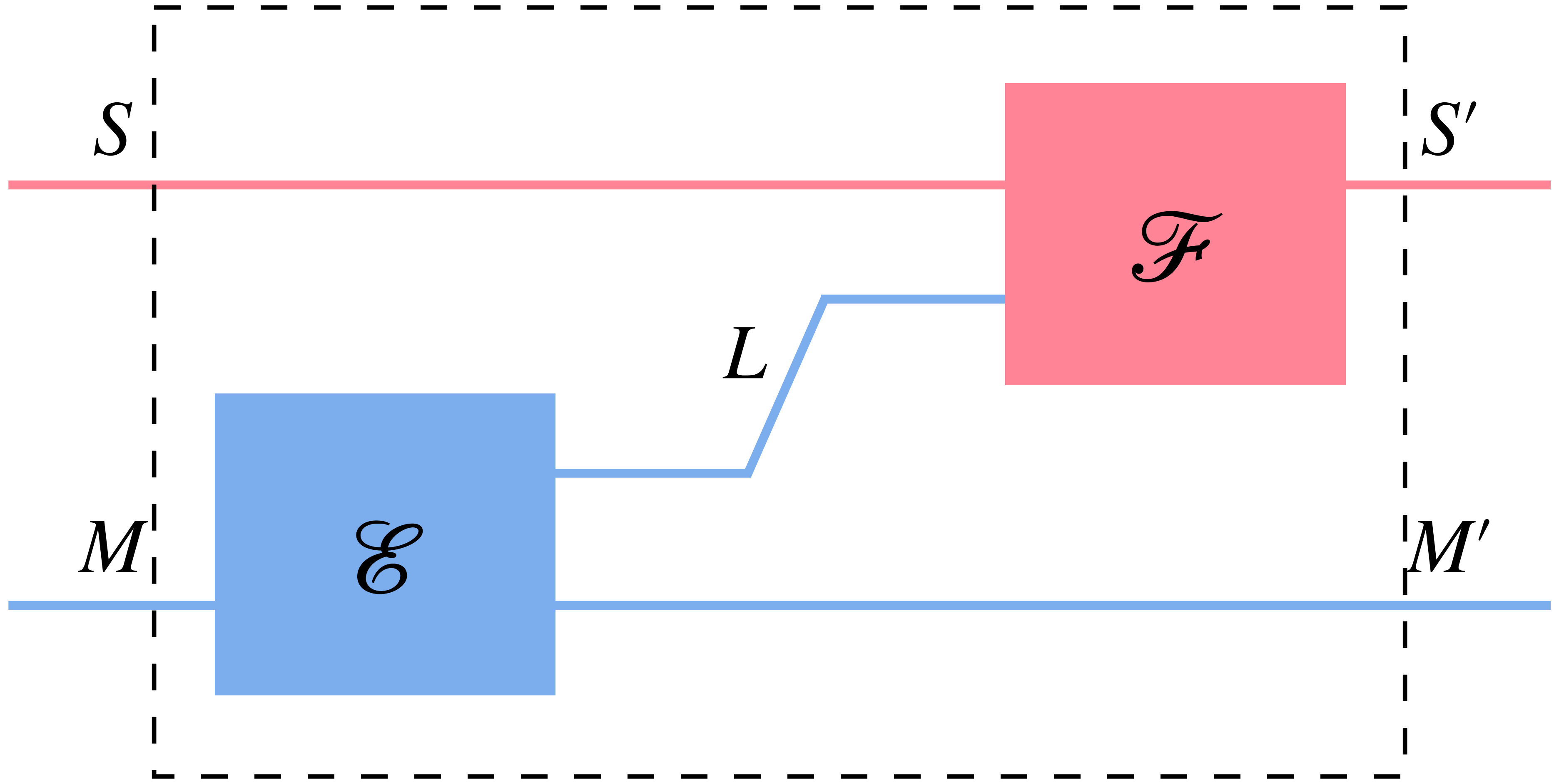

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If something is impossible under QFGOs, then it is impossible via any feedback-control scheme regardless of operational assumptions

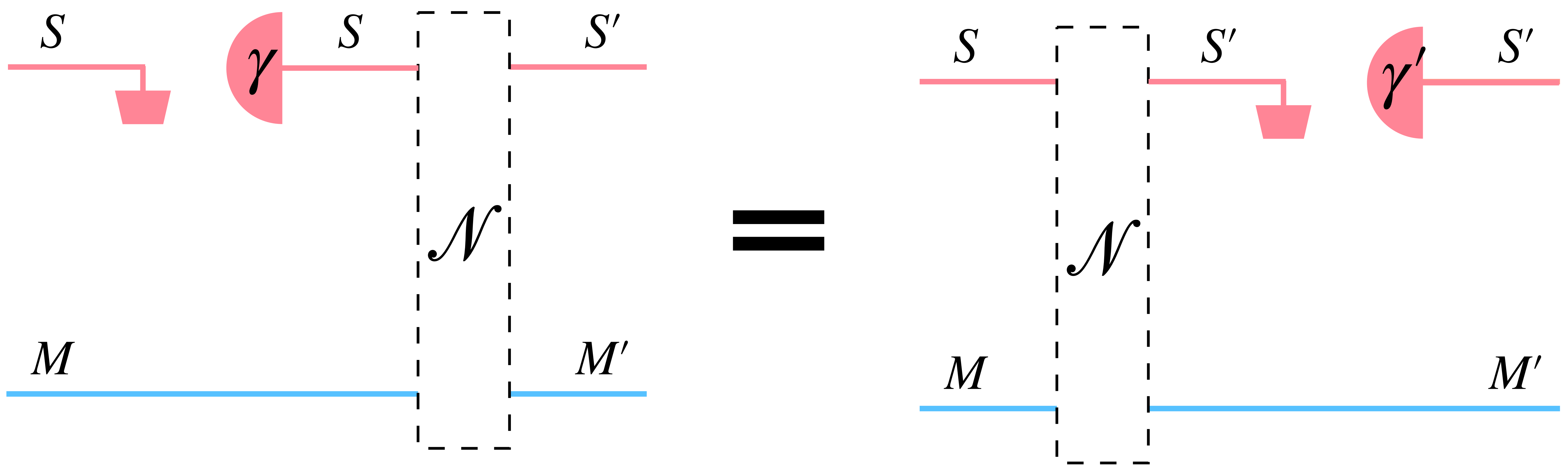
QFGOs: equivalent characterization

Lemma. (Thermalization covariance)

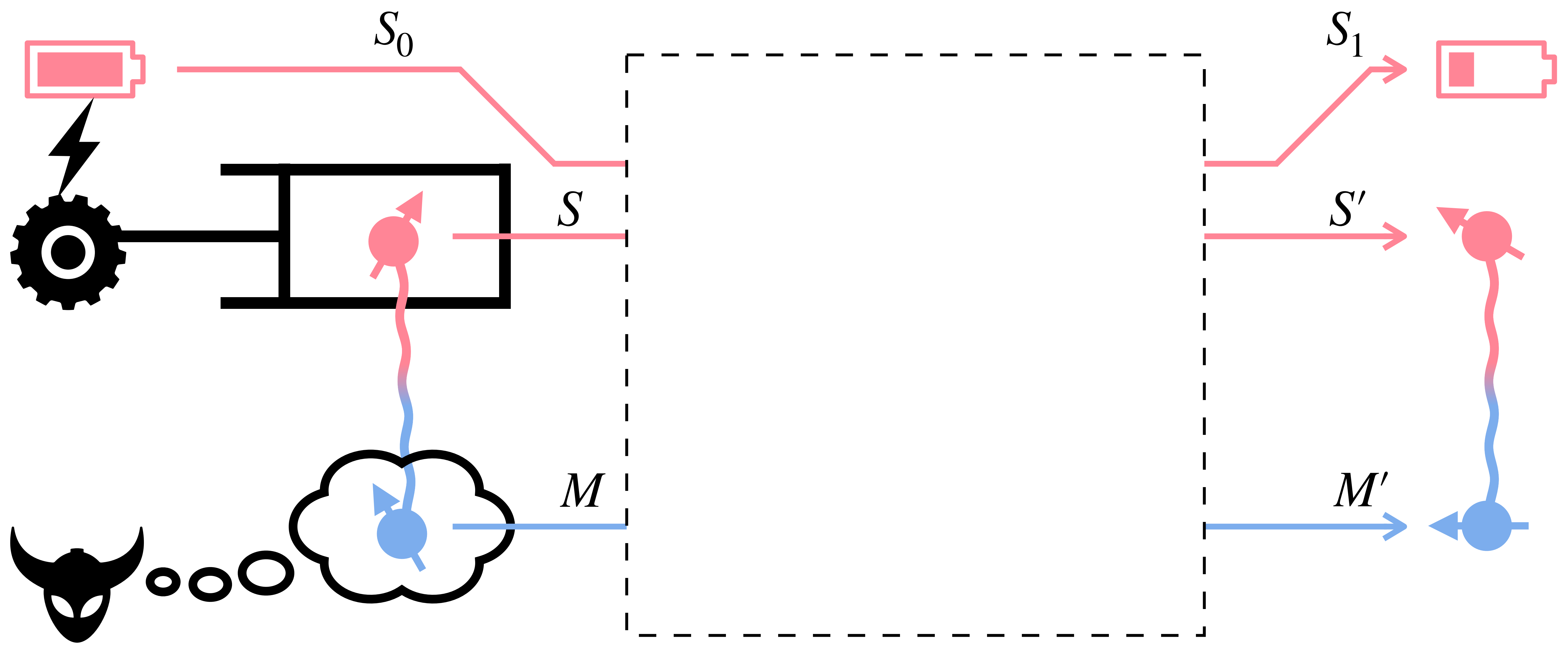
A channel $\mathcal{N}_{SM\to S'M'}$ is a QFGO if and only if

$$\mathcal{N}_{SM\to S'M'} \circ \mathcal{R}^{\gamma}_{S\to S} = \mathcal{R}^{\gamma'}_{S'\to S'} \circ \mathcal{N}_{SM\to S'M'}$$

thermalization of system: $\mathcal{R}^{\gamma}_{S\to S}[\,\cdot\,] = \mathrm{Tr}_S[\,\cdot\,]\gamma_S$



Work accounting



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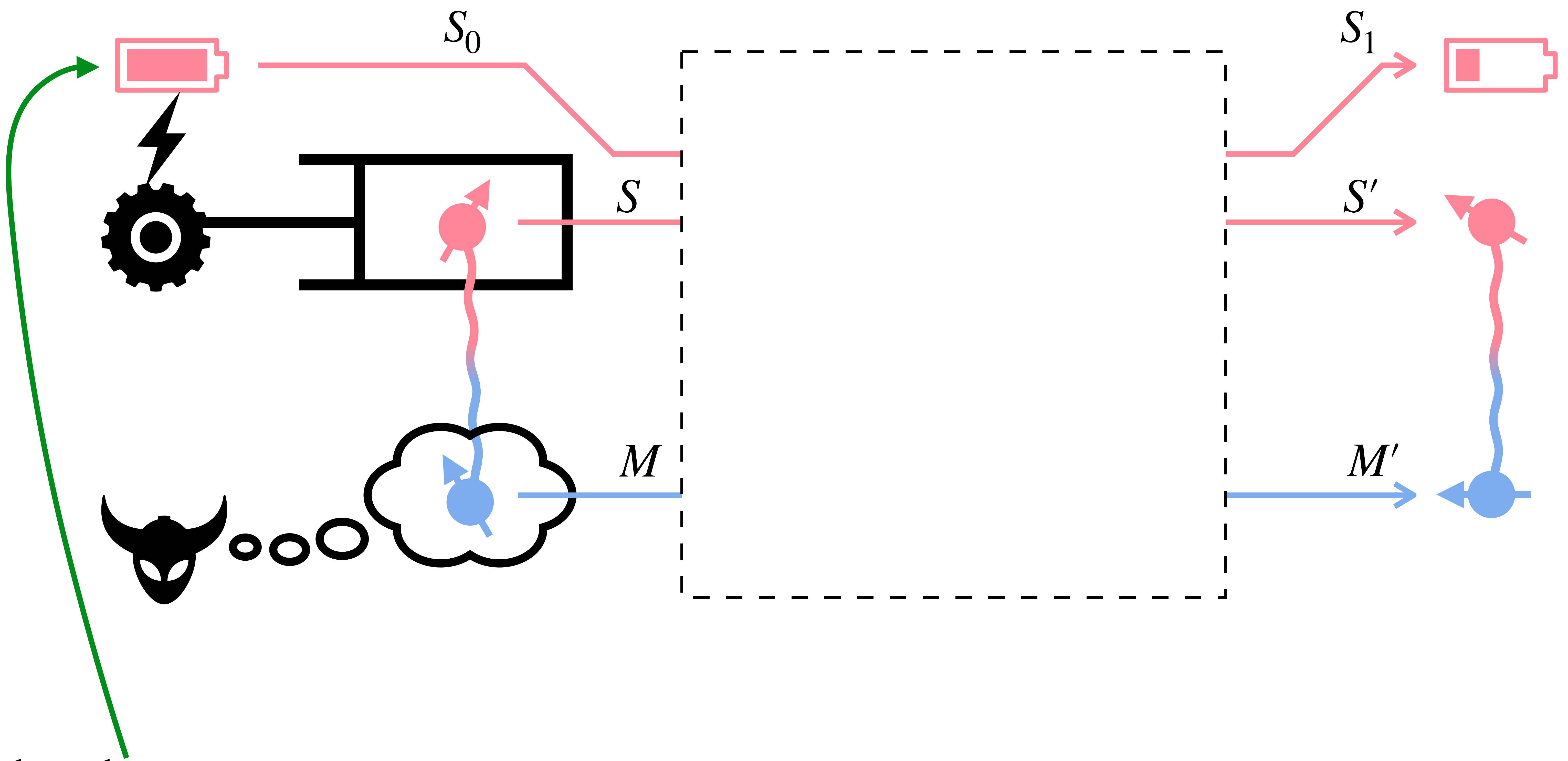
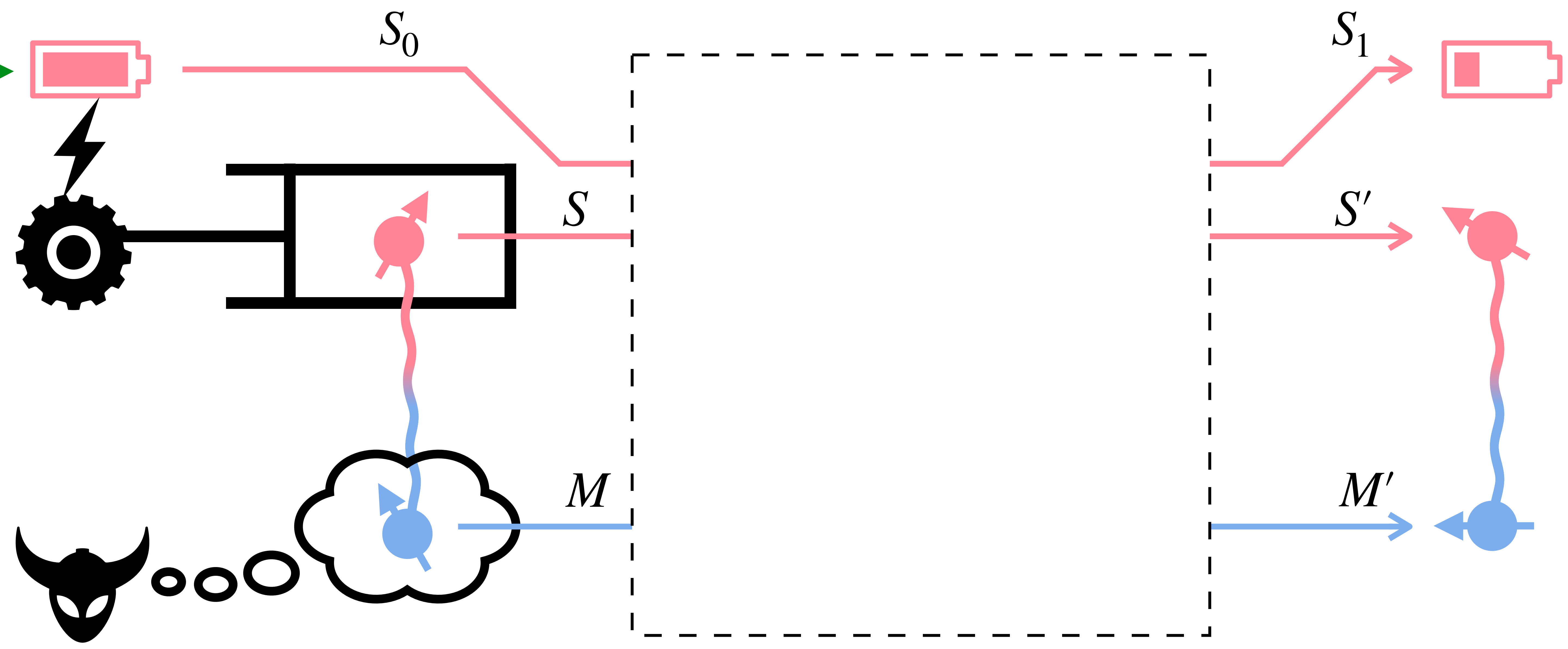


Allow system to exchange work with a battery

Work accounting

- Battery is always in pure state $|0\rangle\langle 0|$ and has uniform Gibbs state
- Battery of dimension d has work storage $\beta^{-1} \ln d$



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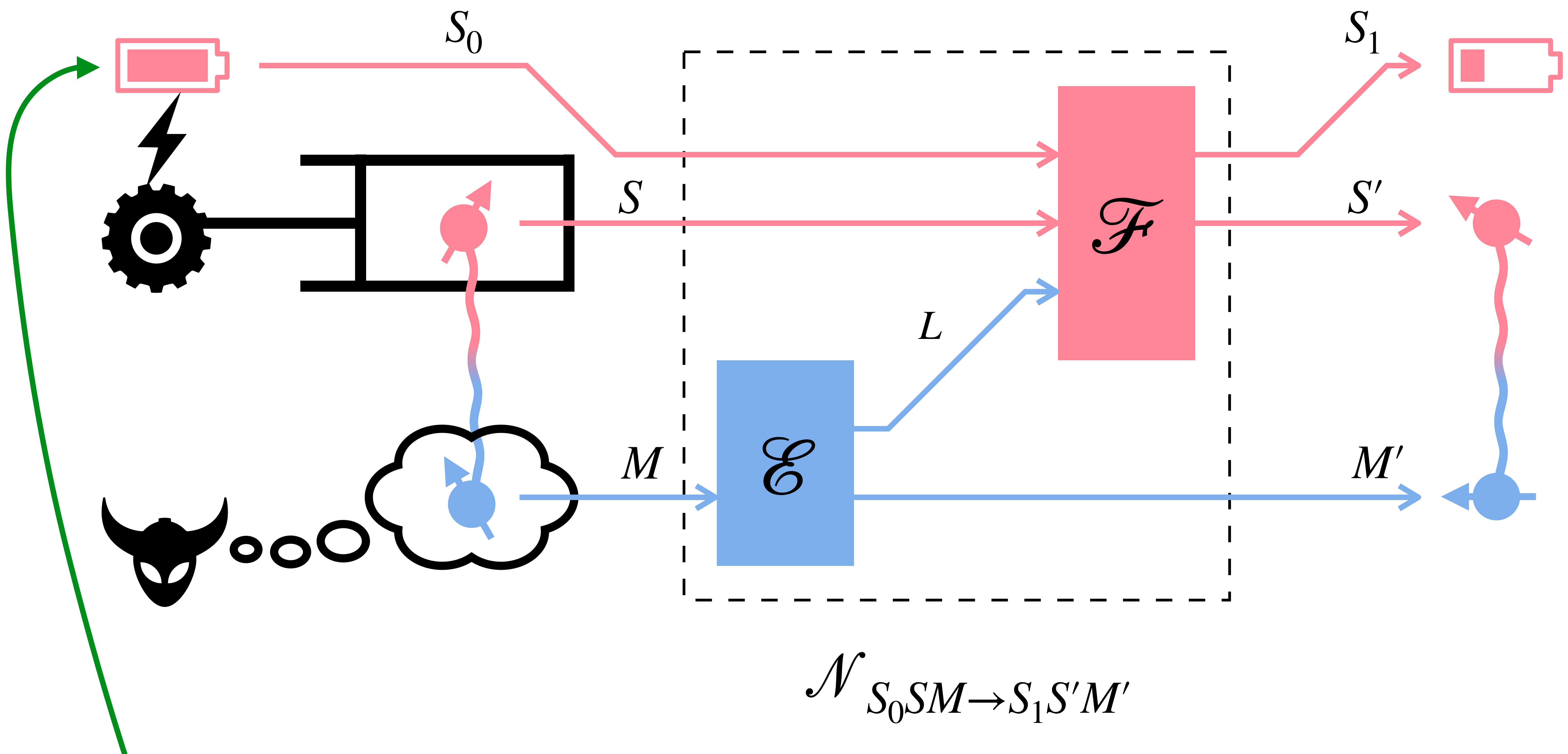

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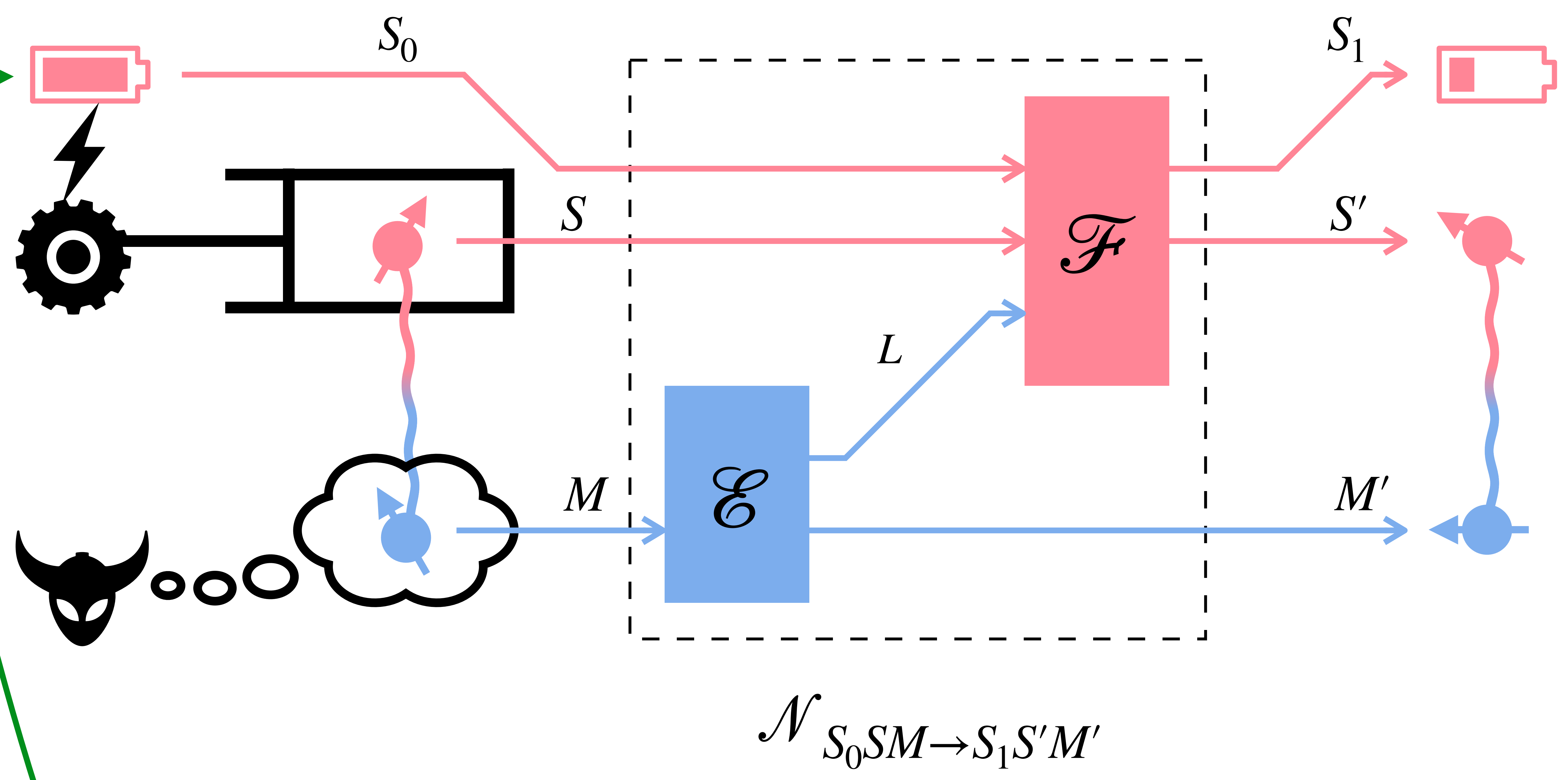


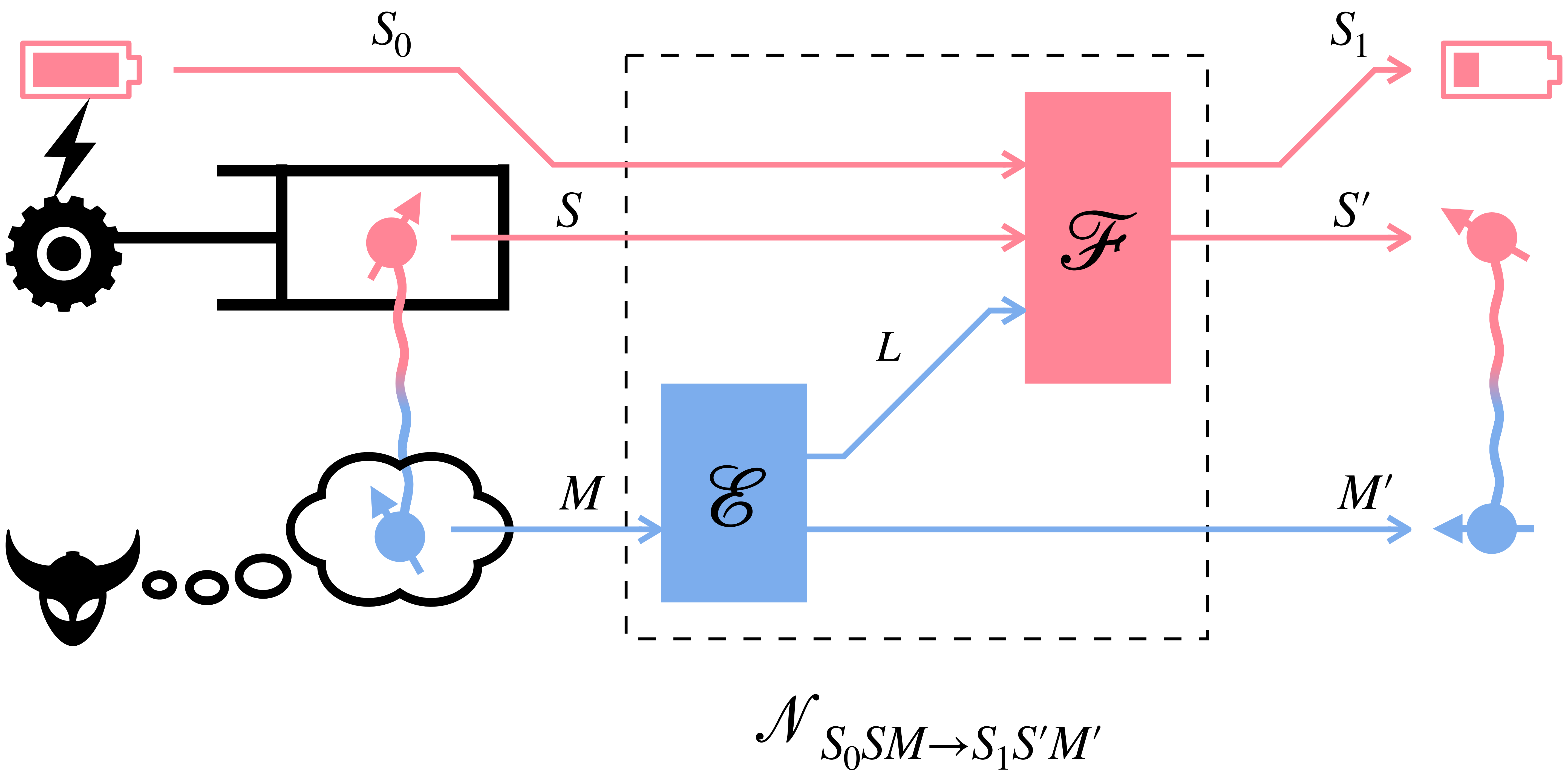
Allow system to exchange work with a battery

Battery-system-memory undergo free operation (QFGO)

By Landauer, the decrease in battery's work storage is work expended on system during conversion: $\beta^{-1}(\ln|S_0| - \ln|S_1|)$

if $|S_1| < |S_0|$, work is consumed; if $|S_1| > |S_0|$, work is gained

Definition of work cost



Single-shot work cost under QFGOs: least amount of work required to convert (ρ_{SM}, γ_S) to $(\rho'_{S'M'}, \gamma'_{S'})$ up to error ε using QFGO

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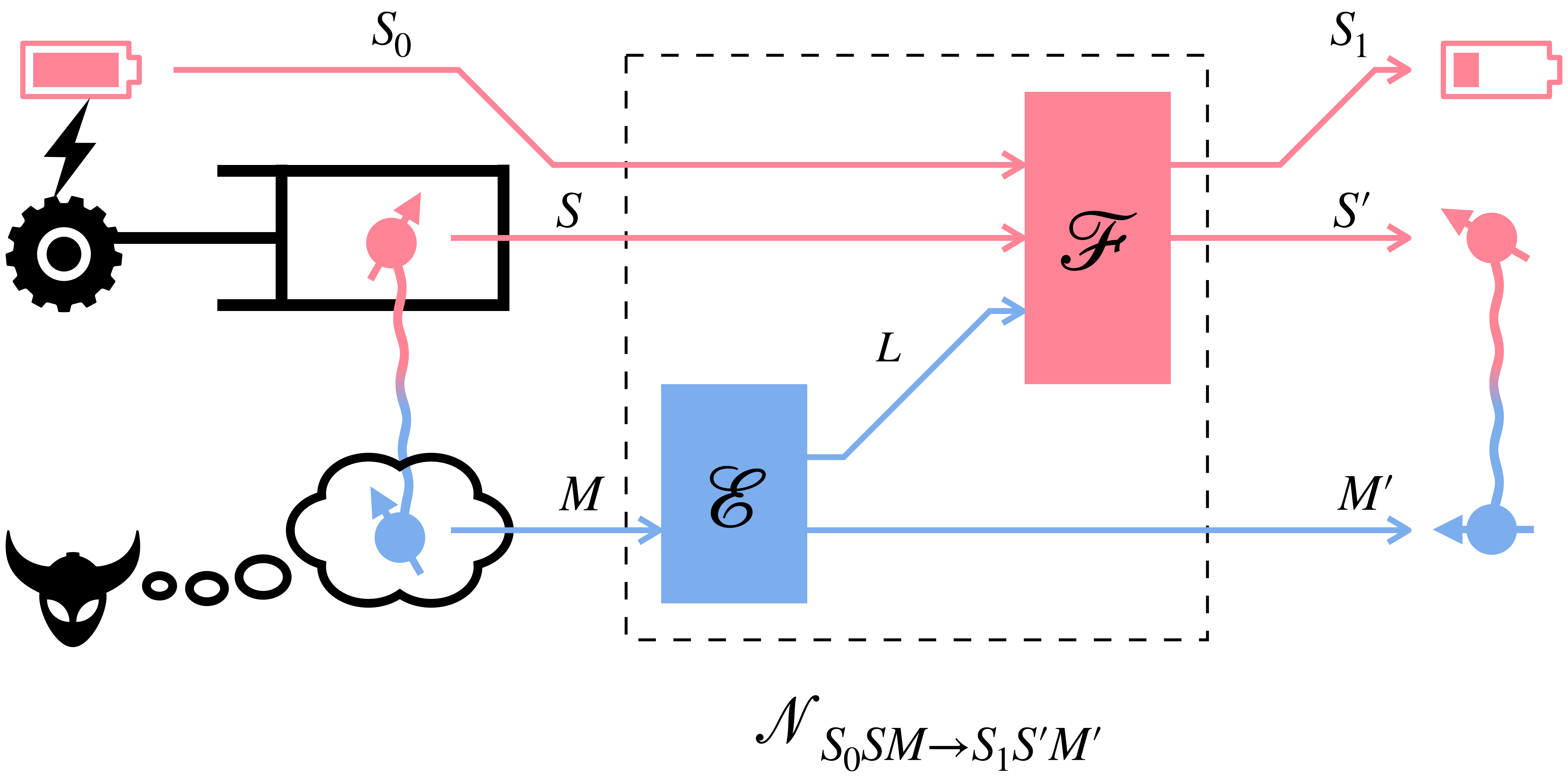

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$$W^{\varepsilon}((\rho_{SM}, \gamma_S) \to (\rho'_{S'M'}, \gamma'_{S'}))$$

in trace distance

$$= \inf_{\substack{S_0, S_1,\\ \mathscr{N} \in \mathrm{QFGO}}} \left\{ \beta^{-1}(\ln|S_0| - \ln|S_1|) : \mathscr{N}_{S_0SM\to S_1S'M'}[\,|0\rangle\langle 0|_{S_0} \otimes \rho_{SM}] \approx_{\varepsilon} |0\rangle\langle 0|_{S_1} \otimes \rho'_{S'M'} \right\}$$

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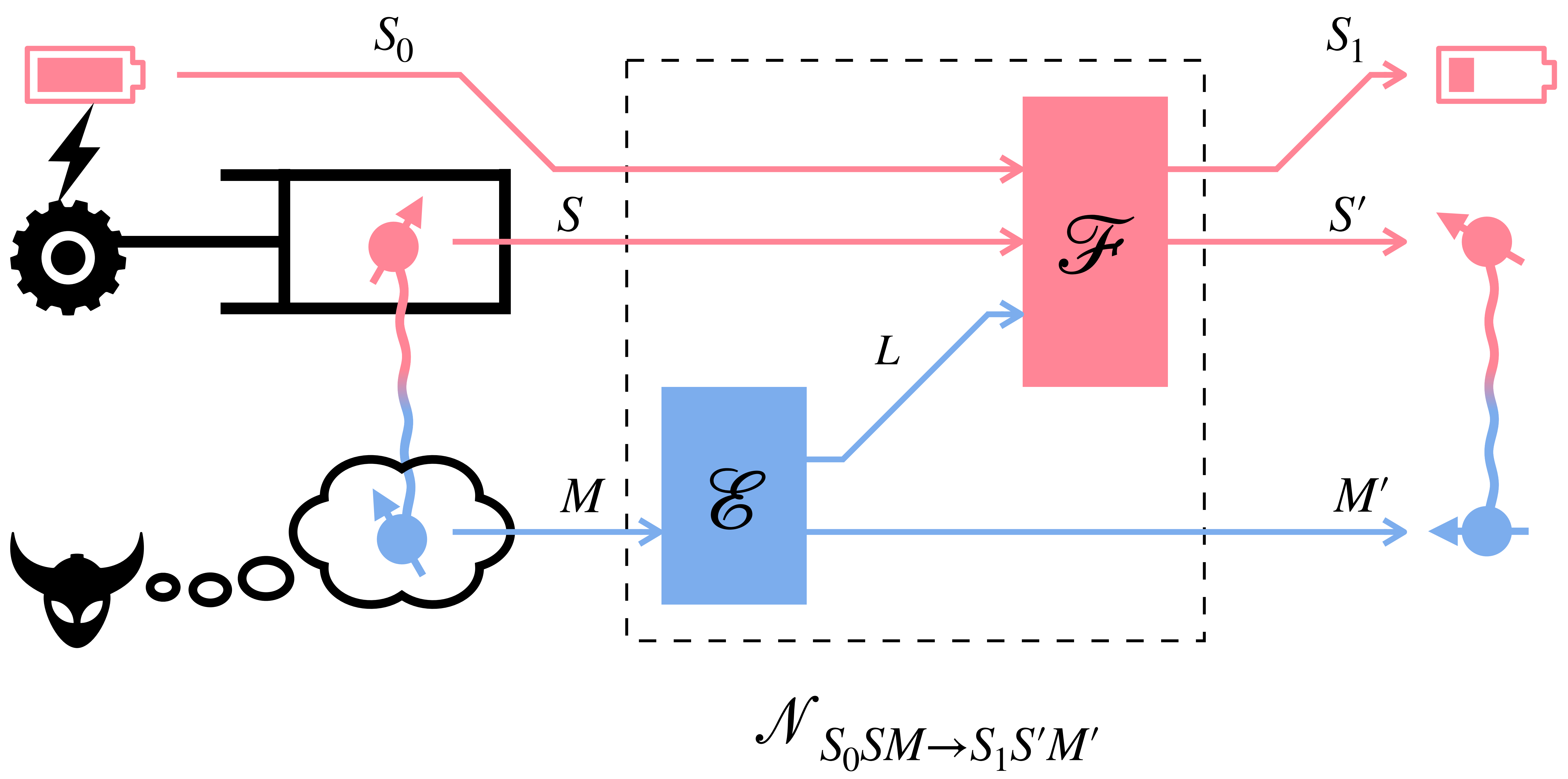

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Single-shot work cost under any feedback-control scheme cannot be smaller than this

Main results:

Single-shot & asymptotic work with q. feedback

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Definition: single-shot work of formation under QFGOs

$$W^{\varepsilon}_{\text{form}}(\rho_{SM}, \gamma_S) := W^{\varepsilon}((\gamma_S, \gamma_S) \to (\rho_{SM}, \gamma_S))$$

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Theorem. Single-shot & asymptotic work of formation under QFGOs:

$$W^{\varepsilon}_{\text{form}}(\rho_{SM},\gamma_S) = \beta^{-1} I^{\uparrow,\varepsilon}_{\max}(\rho_{SM}\|\gamma_S) \qquad \forall\varepsilon\in[0,1]$$

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Smoothed max-generalized mutual information:

$$I^{\uparrow,\varepsilon}_{\max}(\rho_{SM} \| \gamma_S) := \inf_{t, \tau_{SM}, \sigma_M} \{\ln t \colon \tau_{SM} \le t\gamma_S \otimes \sigma_M,\ \tau_M \le \sigma_M,\ \tau_{SM} \approx_\varepsilon \rho_{SM}\}$$

τ_{SM} is subnormalized state, σ_M is normalized state

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a new asymptotic equipartition property

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More on smoothed-max GMI

$\delta(\tau,\rho) := \frac{1}{2}(\|\tau-\rho\|_1 + |\mathrm{Tr}[\tau] - \mathrm{Tr}[\rho]|)$

in generalized trace distance

$$I^{\uparrow,\varepsilon}_{\max}(\rho_{SM}\|\gamma_S) := \inf_{t,\tau_{SM},\sigma_M} \{\ln t \colon \tau_{SM} \le t\gamma_S \otimes \sigma_M,\ \tau_M \le \sigma_M,\ \tau_{SM} \approx_\varepsilon \rho_{SM}\}$$

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- If $\varepsilon = 0$: $I^{\uparrow}_{\max}(\rho_{SM}\|\gamma_S) = D_{\max}(\rho_{SM}\|\gamma_S \otimes \rho_M)$
- If $\varepsilon = 0$ and $\gamma_S = I_S$: $I^{\uparrow}_{\max}(\rho_{SM}\|I_S) = -H^{\downarrow}_{\min}(S|M)_\rho$

unoptimized conditional min-entropy

What's new about it?

- A different kind of smoothing: designed to ensure robustness in the presence of battery:

$$I^{\uparrow,\varepsilon}_{\max}(|0\rangle\langle 0|_{S_1} \otimes \rho_{SM}\|I_{S_1}/|S_1| \otimes \gamma_S) = I^{\uparrow,\varepsilon}_{\max}(\rho_{SM}\|\gamma_S) + \ln|S_1|$$

in standard trace distance

- Alternative expression: $I^{\uparrow,\varepsilon}_{\max}(\rho_{SM}\|\gamma_S) = \inf_{S_1,\omega_{S_1SM}} \{I^{\uparrow}_{\max}(\omega_{S_1SM}\|I_{S_1} \otimes \gamma_S) \colon \omega_{S_1SM} \approx_\varepsilon |0\rangle\langle 0|_{S_1} \otimes \rho_{SM}\}$

normalized state

Proof sketch of single-shot work of formation under QFGOs

Converse: smoothed max-GMI is

(i) resource monotone under QFGOs

(ii) robust in the presence of battery:

$$I_{\max}^{\uparrow,\varepsilon}(|0\rangle\langle 0|_{S_1} \otimes \rho_{SM} \| I_{S_1}/|S_1| \otimes \gamma_S) = \ln|S_1| + I_{\max}^{\uparrow,\varepsilon}(\rho_{SM}\|\gamma_S)$$

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Alternative expression of smoothed max-GMI:

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Achievability: by the following QFGO

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- Saturates the converse:

$$\beta^{-1}(\ln|S_0| - \ln|S_1|) = \beta^{-1}I^{\uparrow}_{\max}(\omega_{S_1SM}\|I_{S_1}/|S_1|\otimes\gamma_S) - \ln|S_1| = \beta^{-1}I^{\uparrow,\varepsilon}_{\max}(\rho_{SM}\|\gamma_S)$$

Task 2: work extraction with q. feedback

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Morris et al., PRL 122, 130601 (2019)

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Definition: single-shot extractable work under QFGOs

$$W^{\varepsilon}_{\text{extr}}(\rho_{SM}, \gamma_S) := -\,W^{\varepsilon}((\rho_{SM}, \gamma_S) \to (\gamma_S, \gamma_S))$$

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Theorem. Single-shot & asymptotic extractable work under QFGOs:

$$W^{\varepsilon}_{\text{extr}}(\rho_{SM}, \gamma_S) = \beta^{-1} I^{\downarrow,\varepsilon}_{\min}(\rho_{SM} \| \gamma_S) \qquad \forall \varepsilon \in [0,1]$$

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$$= \inf_{\sigma_M} D^{\varepsilon}_{\min}(\rho_{SM} \| \gamma_S \otimes \sigma_M)$$

hypothesis-testing relative entropy

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Special case: if $\gamma_S = I_S/|S|$ (trivial Hamiltonian), then single-shot work cost of erasure

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unlike classical case, q. conditional entropies can be negative

cooling instead of heating during erasure in the presence of q. side info.

Proof sketch of single-shot extractable work under QFGOs

Converse: smoothed min-GMI is also

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- Saturates the converse: $\beta^{-1}(\ln|S_1| - \ln|S_0|) = -\beta^{-1}\ln\|\mathrm{Tr}_S[\Lambda_{SM}\gamma_S]\|_\infty = \beta^{-1}I^{\downarrow,\varepsilon}_{\min}(\rho_{SM}\|\gamma_S)$

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Theorem. For a general conversion $(\rho_{SM}, \gamma_S) \to (\rho'_{S'M'}, \gamma'_{S'})$, the average work cost obeys:

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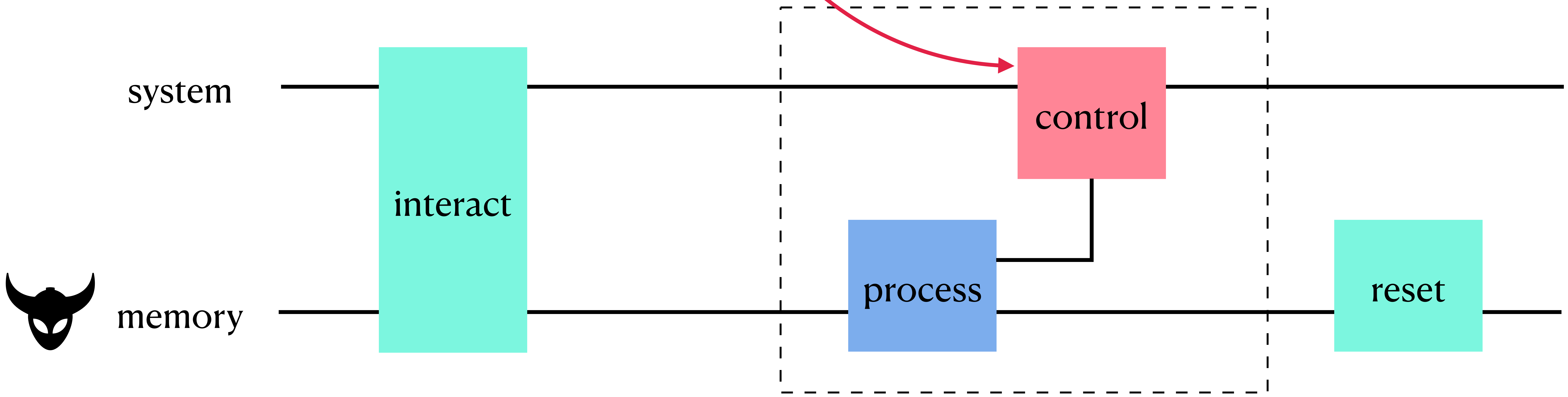

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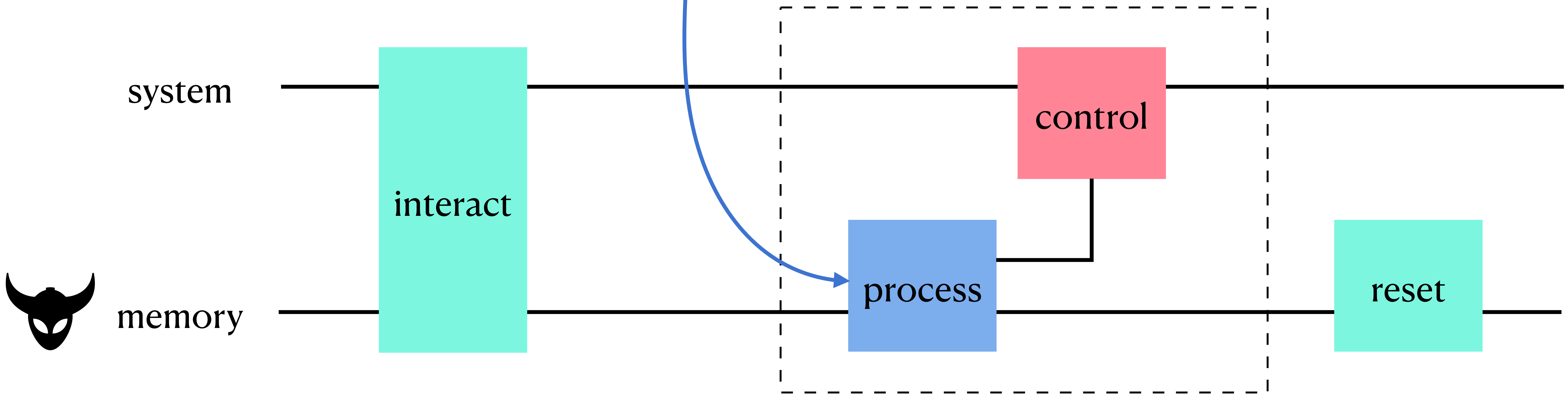

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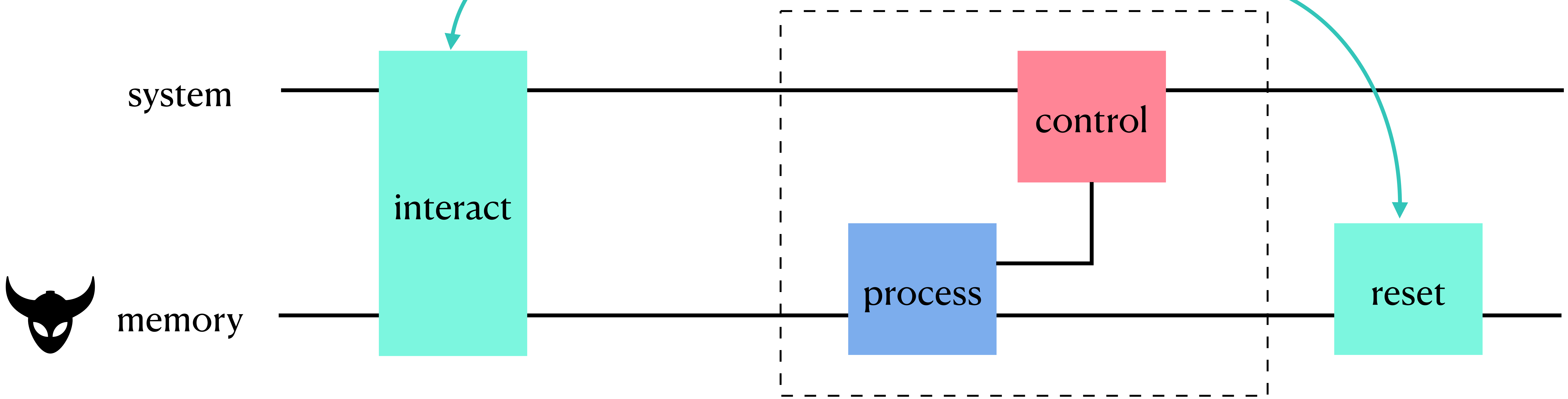

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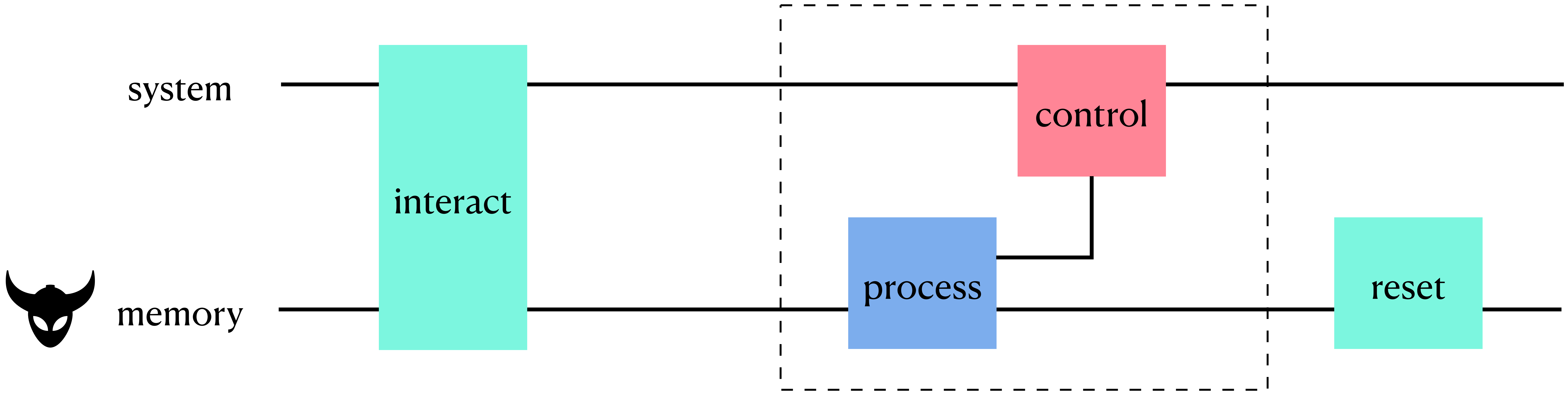

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Gour et al., Quantum 8, 1529 (2024)

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